U.S. Nuclear Regulatory Commission
Site-Specific
Written Examination

Applicant Information

Name:	Region: I
Date: 12/19/2016	Facility: Salem 1 & 2
License Level: RO	Reactor Type: W
Start Time:	Finish Time:

Instructions

Use the answer sheets provided to document your answers. Staple this cover sheet on top of the answer sheets. The passing grade requires a final grade of at least 80.00 percent. Examination papers will be collected SIX hours after the examination starts.

Applicant Certification

All work done on this examination is my own. I have neither given nor received aid.

____
Applicant's Signature

Results

Examination Value	____________ Points
Applicant's Score	____________ Points
Applicant's Grade	____________ Percent

Reactor Operator Answer Key

1.	c	26.	c
2.	b	27.	a
3.	a	28.	a
4.	b	29.	a
5.	d	30.	a
6.	d	31.	c
7.	b	32.	a
8.	d	33.	c
9.	c	34.	a
10.	a	35.	a
11.	a	36.	d
12.	d	37.	d
13.	c	38.	c
14.	d	39.	d
15.	b	40.	b
16.	b	41.	d
17.	b	42.	c
18.	c	43.	b
19.	d	44.	c
20.	b	45.	b
21.	d	46.	d
22.	c	47.	a
23.	d	48.	b
24.	d	49.	a
25.	a	50.	b

Reactor Operator Answer Key

51 . b

52 . d

53 . a

54 . d

55 . d

56 . d

57 . d

58 . a

59 . b

60 . a

61 . a

62 . b

63 . c

64 . a

65 . d

66 . c

67 . c

68 . c

69 . b

70 . d

71 . d

72 . d

73 . a

74 . c

75 . c

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 1

Given the following conditions:

- Unit 1 was operating at 80% power, steady state.
- RCS Cb is 1100 ppm.
- Control rods begin withdrawing at 8 spm.

Of the following, which could be causing this rod withdrawal?

a. 1PT-505, Turbine Steamline Inlet Pressure transmitter, has failed low.

b. 1PT-505, Turbine Steamline Inlet Pressure transmitter, has failed high.

c. A VCT auto makeup occurred, and the 1CV175 Rapid Borate Stop Valve was open.

d. A VCT auto makeup occurred, and the boron addition rate was set to 1 gpm instead of 11 gpm.

Answer c **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000001A202 AA2.02 **RO Value:** 4.2 **SRO Value:** 4.2 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Continuous Rod Withdrawal 001

KA Statement: Ability to determine and interpret the following as they apply to Continuous Rod Withdrawal:
Position of emergency boration valve

Explanation of Answers: 55.41.(b)5 - The system characteristics of the emergency borate line are such that no flow will occur with a BAT pump running in slow speed, which is the normal alignment of the CVCS makeup system. When an auto makeup occurs, the BAT pump running (in auto) swaps to fast speed, and would provide flow through an open CV75 to borate the RCS. The boration would cause RCS temperature to lower, causing SG pressure to lower, causing MT governor valve to open, causing temperature to further lower, and rods would begin withdrawing to reestablish temperature. With power at 80% as stated in the stem, rods would NOT be full out. A is incorrect because PT-505 failing low would cause rod insertion, not withdrawal. B is incorrect because the rod withdrawal would be at maximum rate (72 spm). D is incorrect because it would cause a dilution, not a boration.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Charging, Letdown and Seal Injection	205228-1			74
CVCS Lesson Plan	NOSO5CVCS00-16			16

L.O. Number
ABROD3E001

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 2

With Unit 2 at power, which of the following failures would cause a **SINGLE** control rod to drop fully into the core?

a. Blown fuse for a Lift coil during demand for rod movement.

b. Blown fuse for a Stationary Gripper coil with no demand for rod movement.

c. Loss of Power to a single Power Cabinet during demand for rod movement.

d. Loss of Power to a single Logic Cabinet with no demand for rod movement.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000003K205 AK2.05 **RO Value:** 2.5 **SRO Value:** 2.8 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Dropped Control Rod 003

KA Statement: Knowledge of the interrelations between Dropped Control Rod and the following:
Control rod drive power supplies and logic circuits

Explanation of Answers: 55.41.(b)6- Rod movement loss of lift coil still has Moveable Gripper holding rod, rod just wouldn't move. Stationary gripper coil holds rod while not moving. Loss of power to Power Cabinet would affect multiple rods stationary gripper coils. Loss of Logic cabinet with a demand for rod movement could cause rods to drop during the movement cycle, as it would lose power to movable gripper and lift coils, and rods would stop when cycle timed out and stationary gripper coils re-engaged.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Rod Control Lesson Plan	NOS05RODS-12		21,44,55,	12

L.O. Number
RODS00E007

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

Question Topic RO 3

Given the following conditions:

- Unit 1 was operating at 100% power when a reactor trip signal was generated by the RPS, but the Reactor Trip Breakers did NOT open.
- The reactor was tripped from the control room when the RDMG sets power supply breakers were opened from 1CC3.

Because RPS did not successfully trip the reactor, which of the following actions is required to be directly performed by the control room crew?

a. Trip the Main Turbine.

b. Start the MDAFW pumps.

c. Transfer 4KV Group buses.

d. Open the Main Generator Output Breakers.

Answer a **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000007K202 EK2.02 **RO Value:** 2.6 **SRO Value:** 2.8 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Reactor Trip 007

KA Statement: Knowledge of the interrelations between Reactor Trip and the following:
Breakers, relays and disconnects

Explanation of Answers: 55.41(b)7 - With the reactor trip breakers failing to open, the Main Turbine will not receive a trip signal (from the output of the RTBs). A manual turbine trip will be performed. The main turbine trip will cause the main generator output breakers to automatically open, so the crew will not directly open the output breakers. The group buses will transfer automatically on low voltage. The AFW pumps will automatically start on lo-lo level when the SGs shrink after the main turbine is tripped.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Reactor Protection System	221051			13

L.O. Number | **Objectives**

TRP001E021

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 4

What does EOP-LOCA-1, Loss of Reactor Coolant, check for OTHER than a LOCA as the reason for being in LOCA-1, and why?

a. SGTR. If the ECCS injection flow is due to a SGTR, the tube rupture must be addressed before returning to LOCA-1 to terminate ECCS flow.

b. Steam break. If the ECCS injection flow is due to a Loss of Secondary Coolant, the event could be terminated by isolating the faulted SG.

c. SGTR. If the ECCS injection flow is due to a SGTR, an unnecessary transition to LOCA-3, Transfer to Cold Leg Recirculation would be made.

d. Steam break. If the ECCS injection flow is due to a Loss of Secondary Coolant, an unnecessary transition to FRHS-1, Loss of Secondary Heat Sink could be required.

Answer	b	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	000009G418	2.4.18	**RO Value:**	3.3	**SRO Value:**	4.0	**Section:**	EPE	**RO Group:**	1	**SRO Group:**	1	55.43

System/Evolution Title Small Break LOCA 009

KA Statement: Knowledge of the specific bases for EOPs.

Explanation of Answers: 55.41.b(10)The Major Action Categories for LOCA-1 are : 1. Check for Subsequent Failure, 2. Monitor Plant Equipment for Optimal Mode of Operation, and 3. Determine optimal Method of Long-Term Plant Recovery. The first item checks that a faulted or ruptured SG is not the reason for ECCS injection, and either fixes it on the spot (faulted) or transitions to another more appropriate procedure With a faulted SG being the cause of the ECCS flow, LOCA-1 has a "do loop" which will wait until the SG has blown down to go to TRIP-3 SI Termination, vs staying in LOCA-1 until the transition to LOCA-2 is made. A transition to SGTR is performed to do actions which will terminate the primary to secondary leakage, which is not performed in LOCA-1. Additionally, there is no transition from SGTR-1 to LOCA-1. The unnecessary transition to FRHS-1 is plausible because it happens on other procedures (FRCE,LOSC-2) when minimizing AFW flow to SGs when all are faulted.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Reactor Coolant	1-EOP-LOCA-1			30

L.O. Number | Objectives

LOCA01E009

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 5

Given the following conditions:

- Due to a LBLOCA, Operators have entered 2-EOP-LOCA-3, Transfer to Cold Leg Recirculation.
- NO CCW pumps are running.

Which of the following describes the procedure flowpath?

a. Continue in LOCA-3 to align BOTH RHR pumps, one SI pump, and one charging pump for recirc until RWST lo-lo level is reached.

b. Transition to LOCA-5, Loss of Emergency Coolant Recirculation, while attempting to restore CCW.

c. Transition to APPX-1, CCW Restoration, then establish cold leg recirculation.

d. Continue in LOCA-3 and align for single train recirculation without CCW.

Answer d **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000011A203 EA2.03 **RO Value:** 3.7 **SRO Value:** 4.2 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Large Break LOCA 011

KA Statement: Ability to determine and interpret the following as they apply to Large Break LOCA:
Consequences of managing LOCA with loss of CCW

Explanation of Answers: 55.41.b(10) At LOCA-3 step 11.2, with no CCW pumps running, operators are directed to go to step 124 and align for single train operation. There is no provision to transition to APPX-1 or LOCA-5.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Transfer to Cold Leg Recirculation	2-EOP-LOCA-3			30

L.O. Number

LCA3U2E006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program**

Question Source Comments 74865

Comment

RO SkyScraper	SRO Skyscraper	RO System/Evolution List	SRO System/Evolution List	Outline Changes

Question Topic RO 6

Which of the following indications would be **unexpected** following a total failure of a RCP #1 seal?

Affected RCP.....

a. Seal leakoff D/P low console alarm.

b. Standpipe level high console alarm.

c. #1 seal outlet temperature rising.

d. Seal flow lowering.

Answer	d	Exam Level	R	Cognitive Level	Application	Facility:	Salem 1 & 2	ExamDate:	12/19/2016

KA:	000015K210	AK2.10	RO Value:	2.8*	SRO Value:	2.8	Section:	EPE	RO Group:	1	SRO Group:	1	55.43

System/Evolution Title Reactor Coolant Pump Malfunctions 015

KA Statement: Knowledge of the interrelations between Reactor Coolant Pump Malfunctions and the following:
RCP indicators and controls

Explanation of Answers: 55.41.b(3)Number 1 seal failure means minimum resistance / maximum seal flow, so seal flow will rise. More seal flow will be felt at the #2 seal, which will cause higher #2 seal D/P and higher leakoff going to the standpipe. The seal leakoff D/P alarm will come in because the failure of the seal results in no D/P across the seal.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
RCP Lesson Plan	NOS05RCPUMP-12		62	12

L.O. Number
RCPUMPE016

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 7

Given the following conditions:

- Unit 2 was operating at 90% power with 21 charging pump in service when the controlling PZR Level Channel I failed low.
- The Charging Master Flow Controller was placed in Manual when directed by S2.OP-AB.CVC-0001, Loss of Charging.
- The alternate PZR level channel has been selected as the controlling channel.
- Letdown has been restored.

Which of the following identifies a consequence of returning the Master Flow Controller to auto PRIOR to returning PZR level to program as directed in S2.OP-AB.CVC-0001 ?

Charging flow will...

a. rise, and VCT auto makeup may initiate.

b. lower, and flashing in the Letdown line could occur.

c. rise, and RCP seal injection flow could exceed Tech Spec limit total seal injection flow.

d. lower, and 2CC71 LTDWN HX CC CONT VALVE will not respond quickly enough to prevent Mixed Bed Demin isolation on high inlet temperature

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000022K103 AK1.03 **RO Value:** 3.0 **SRO Value:** 3.4 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Loss of Reactor Coolant Makeup 022

KA Statement: Knowledge of the operational implications of the following concepts as they apply to Loss of Reactor Coolant Makeup: Relationship between charging flow and PZR level

Explanation of Answers: 55.41.b(7) With a CCP in service, the failure LOW of the controlling PZR level channel will cause charging flow to RISE. The stem stated that MFC was taken to manual when directed IAW AB, so there was sufficient time for actual charging flow to rise substantially. With actual level higher than programmed level, if the MFC were placed in auto it would force charging flow to lower. If charging flow lowered to <~60 gpm, inadequate cooling of letdown flow would occur in the regenerative heat exchanger, and letdown line flashing would occur. The CC71 is normally only ~10% open, and has plenty of room to open if letdown temp were to rise downstream of the letdown HX, and temps would not reach demin isolation levels. The 2 rises are incorrect because charging flow wouldn't rise, but the actions associated with higher flow are correct.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Charging	S2.OP-AB.CVC-0001			9

L.O. Number
ABCVC1E002

Objectives

Material Required for Examination	

Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐

Question Source Comments	125676

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 8

When responding to multiple stuck rods during the performance of EOP-TRIP-2, Reactor Trip Response, why are there two different boration times associated with borating the RCS for those stuck rods?

a. As the number of stuck out rods increases, the amount of boration required to be added per rod also increases.

b. If a cooldown is in progress a higher boration total is required vs if the plant is stable in Hot Standby.

c. The time in core life (BOL vs EOL) require different specific boration times to be used.

d. The different available borated water sources contain different ppm boron.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000024K302 AK3.02 **RO Value:** 4.2 **SRO Value:** 4.4 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Emergency Boration 024

KA Statement: Knowledge of the reasons for the following responses as they apply to Emergency Boration:
Actions contained in EOP for emergency boration

Explanation of Answers: 55.41.b(10) Step 7 of TRIP-2 requires boration for 2 or more control rods not fully inserted. It first attempts rapid boration from the Boric Acid Storage Tanks via a Boric Acid Transfer pump in high speed, for 35 minutes per rod. If this is unsuccessful, then boration from the Refueling Water Storage Tank via a charging pump running at greater than 87 gpm is performed for 120 minutes per rod. The BAST minimum Cb is 6550 ppm per Tech Specs, while the RWST minimum Cb is 2300 ppm.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Reactor Trip Response	2-EOP-TRIP-2	Step 7		30
Boration Times for Multiple Stuck-out Control Ro	S-C-CVC-MDC-1314		15	1

L.O. Number
TRP002E006

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: RO 9

Given the following conditions:

- Unit 2 is in MODE 4 with RHR system providing the shutdown cooling.
- 2RH25 RHR Hot Leg Recirc discharge relief starts to discharge (lift).

With NO operator action, which of the following indications would occur if the valve continues to pass flow?

a. OHA-41 AUX ALM SYS PRINTER, for point 758, 24 RHR SUMP PUMP START.

b. OHA-41 AUX ALM SYS PRINTER, for point 710, RCDT PUMP START.

c. OHA C-2, CNTMT SUMP PUMP START.

d. PRT LEVEL HI-LO Console Alarm.

Answer	Exam Level	Cognitive Level	Facility:	ExamDate:
c	R	Application	Salem 1 & 2	12/19/2016

KA:		RO Value:	SRO Value:	Section:	RO Group:	SRO Group:	
000025G446	2.4.46	4.2	4.2	EPE	1	1	55.43

System/Evolution Title: Loss of Residual Heat Removal System | 025

KA Statement: Ability to verify that the alarms are consistent with the plant conditions.

Explanation of Answers: 55.41.b(5,7,8) The 2RH25 is located upstream of normally shut 2RH26 which supplies Hot Leg Recirc during accident conditions. It is located in containment, and as such, discharges to the containment trench, which flows into the containment pocket sump. C-2 will annunciate when the containment sump pump starts. All the other choices are radioactive waste paths which do not receive the discharge. RCDT and PRT are also located in containment and are plausible relief path collection points. RHR sump pump run would occur if the relief was directed to RHR sump.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Residual Heat Removal	205332-1			

L.O. Number

RHR000E008

Objectives

Material Required for Examination:

Question Source: Facility Exam Bank | **Question Modification Method:** Editorially Modified | **Used During Training Program**

Question Source Comments: 42333 changed from levels changing to alarms which would occur.

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 10

Given the following conditions:

- Unit 2 initiates a Safety Injection from 100% power due to a LOCA.
- 15 minutes after the SI is initiated, containment pressure peaks at 5 psig.

Which of the following identifies an automatic response which has occurred for Component Cooling Water System components, and why?

a. 2CC215 and 2CC113, Excess Letdown Heat Exchanger CCW isolation valves, received a close signal to ensure this non-essential containment flow path is isolated.

b. 21 and 22CC16, RHR HX CCW isolation valves, received an open signal to ensure that long term cooling of the RCS is in service when the swap to Cold Leg Recirc is required.

c. 2CC215 and 2CC113, Excess Letdown Heat Exchanger CCW isolation valves, received a close signal to ensure ALL CCW supply and return from the containment is isolated.

d. 21 and 22CC16, RHR HX CCW isolation valves, received an open signal to ensure that RHR pumps do not overheat if the RCS remains above the shutoff head of the RHR pumps.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000026K302 AK3.02 **RO Value:** 3.6 **SRO Value:** 3.9 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Loss of Component Cooling Water 026

KA Statement: Knowledge of the reasons for the following responses as they apply to Loss of Component Cooling Water:
The automatic actions (alignments) within the CCWS resulting from the actuation of the ESFAS

Explanation of Answers: 55.41.b(7,8) The SI signal sends a close signal to the CC215 and CC113, Excess Letdown HX isolation valves, as they are Containment Phase A isolation valves. The purpose of closing Phase A isolation valves is to ensure all non-essential containment penetrations are isolated on a SI. B is incorrect because CC16s do not receive an open signal until the ARM PB is depressed and the RWST level is 15.2' and manual alignment is required to place ECCS in CLR. C is incorrect because ALL CCW supply and return are not isolated on a SI signal, the RCP CCW is still being supplied until a Phase B signal at 15 psig in cont. D is incorrect because the CC16s do not receive an open signal until the ARM PB is depressed and the RWST level is 15.2', and the RHR pumps are cooled by either flow through the pump from RWST (LBLOCA) or recirc flow (SBLOCA until pp is S/D).

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Component Cooling Lesson Plan	NOS05CCW000-11		42	11

L.O. Number | Objectives

CCW000E004

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 42744

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 11

When responding to lowering PZR pressure IAW S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction, why are BOTH PZR PORV Stop valves closed when attempting to determine if the PORVs are the source of the pressure reduction?

a. Because the PORV tailpipe temperature sensor is on the combined PORV discharge line.

b. To prevent pressurizing the PRT past the point where venting will be unavailable due to the high pressure interlock.

c. Because the PZR PORV and PZR Safety Valves discharge into a single line, and isolation of both PORVs is necessary to determine whether the leak is from the PORV's or Safeties.

d. Because the PZR PORV and Rx Head Vent Valves discharge into a single line, and isolation of both PORVs will give indication of whether the leak is from the PORV's or Rx Head Vents.

Answer	a	Exam Level	R	Cognitive Level	Memory	Facility:	Salem 1 & 2	ExamDate:	12/19/2016

KA:	000027K303	AK3.03	RO Value:	3.7	SRO Value:	4.1	Section:	EPE	RO Group:	1	SRO Group:	1	55.43

System/Evolution Title Pressurizer Pressure Control Malfunction 027

KA Statement: Knowledge of the reasons for the following responses as they apply to Pressurizer Pressure Control Malfunction:
Actions contained in EOP for PZR PCS malfunction

Explanation of Answers: 55.41.b(7) As shown on dwg 203301, both the PZR PORVs and Safeties discharges combine into a single line going to the PRT. The Safeties each have their own individual temperature sensor, while the PORVs have one sensor on a common line. Since there is no way to distinguish which valve is leaking, both PORV Stop valves are closed. If temperature does not lower, the user is directed to reopen the stop valves and go to steps to check safety valve tailpipe temperatures. If the temperature does lower, the user is directed to reopen one PORV Stop (2PR6). If temperature does not rise, then the procedure is exited. If it does, then that stop valve is closed and the other opened. Then the procedure is exited." B is incorrect but plausible because discharge is directed to the PRT, and PRT vent valve is prevented from opening >10 psig. C is incorrect but plausible because the all the discharges do combine into a single line, however all 3 safeties have their own sensors. C is incorrect but plausible because the all the discharges do combine into a single line, however, Rx Head Vent leakage is not addressed in AB.PZR-1, as the vents would be seen as a RCS leak.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Pressurizer Pressure Malfunction	S2.OP-AB.PZR-0001			18

L.O. Number
ABPZR1E002

Objectives

Material Required for Examination	

Question Source:	New	Question Modification Method:		Used During Training Program	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 12

Analyze the following conditions:

- Unit 2 is operating at 100% power.
- I&C is performing Solid State Protection System testing.
- Reactor Trip Breaker (RTB) **A** is open.
- Reactor Trip Bypass Breaker **A** is racked in and closed.
- A valid reactor trip signal is generated in the Reactor Protection System, but the RPS system does not provide any output signal to trip the Rx.

Assuming that the RPS system WOULD generate the appropriate signal when MANUAL actions are performed during TRIP-1 immediate actions, which of the following contains both the operator action and the DIRECT method which will trip the Rx?

a. Depressing the OPEN PB for Reactor Trip Breaker B to deenergize the UV coil.

b. Deenergizing the RDMG sets by opening the PZR heater bus supply breakers on 2CC3.

c. Depressing the OPEN PB for Reactor Trip Bypass Breaker A to energize the shunt coil.

d. Turning either of the Reactor Trip pistol grip handles to the TRIP position to deenergize the UV coil for Reactor Trip Breaker B.

Answer d **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000029K206 AK2.06 **RO Value:** 2.9* **SRO Value:** 3.1* **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Anticipated Transient Without Scram 029

KA Statement: Knowledge of the interrelations between Anticipated Transient Without Scram and the following:
Breakers, relays, and disconnects

Explanation of Answers: 55.41.B(7)he failure of the RPS output signal from the automatic trip will initially prevent the reactor from tripping. A is incorrect because PB energizes the shunt coil. B is incorrect because the Rx would already have tripped by another means prior to having to perform this step.. C is incorrect because the RT Bypass Breakers do not have an OPERATE function from the control console.
Tier/Group 2/1
10CFR55.41-6

Cognitive Level 3 - Analysis
References: NOS05RODS00 ELO-19,
Logic drawing 221051
Schematic drawing 240111

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
RPS Trip Signals	221051			13

L.O. Number
RXPROTE027

Objectives

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments	50753				

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 13

Given the following conditions:

- Unit 1 is operating at 100% power when Intermediate Range Nuclear Instrument 1N35 fails high.
- Operators are responding IAW S1.OP-AB.NIS-0001, Nuclear Instrumentation System Malfunction, and are removing the channel from service IAW S1.OP-SO.RPS-0001, Nuclear Instrumentation Channel Trip / Restoration.
- With the PO in the rack area, OHA E-29, SR & IR TRIP BYP, annunciates.

Which of the following identifies the cause of this alarm?

a. 1N35 Control Power fuses have been removed.

b. 1N35 Instrument Power fuses have been removed.

c. 1N35 LEVEL TRIP switch has been placed in the bypass position.

d. POWER MISMATCH BYPASS switch has been placed in BYPASS 1N35.

Answer c **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000033A102 AA1.02 **RO Value:** 3.0 **SRO Value:** 3.1 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Loss of Intermediate Range Nuclear Instrumentation 033

KA Statement: Ability to operate and / or monitor the following as they apply to Loss of Intermediate Range Nuclear Instrumentation:
Level trip bypass

Explanation of Answers: 55.41.b(7) A is incorrect but plausible since control power fuses are removed when removing a PR NI from service using the same procedure. B is incorrect but plausible as it is the next action to be performed after placing the Level Trip Bypass switch in Bypass. C is correct because when placing the level trip bypass switch the BYPASS position, the IR FLUX HI reactor trip and IR HI FLUX ROD WDRWL STOP are blocked, and OHA E-29 annunciates. D is incorrect because the IR does not have a power mismatch bypass switch, but the PRNI detectors do.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Nuclear Instrumentation Channel Trip / Restorati	S1.OP-SO.RPS-0001		11	5

L.O. Number

ABNIS1E001

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 14

Which of the radiation monitors listed below would provide the FIRST indication to the control room crew that a Steam Generator tube leak is occurring with the unit at 100% power?

a. R15 Condenser Air Ejector Monitor.

b. R19 SG Blowdown Rad Monitor.

c. R46 Main Steamline Monitor.

d. R53 Main Steamline Monitor.

Answer	d	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016				
KA:	000037A106	AA1.06	**RO Value:**	3.8*	**SRO Value:**	3.9*	**Section:**	EPE	**RO Group:**	2	**SRO Group:**	2	55.43

System/Evolution Title Steam Generator Tube Leak 037

KA Statement: Ability to operate and / or monitor the following as they apply to Steam Generator Tube Leak:
Main steam line rad monitor meters

Explanation of Answers: 55.41.b(11) All the listed monitors will show some indication of a SGTL. However, the R53 are N2 monitors in the Main Steamline, which are very sensitive and would indicate prior to the others. The R15 samples the Main Condenser, so the steam would have to get there first (past the R53 monitors) The r46 are high range monitors, and while they can detect low levels of radiation, they would not provide any indication to control room crew via an alarm to alert them to monitor them. The R19 blowdown monitor would provide indication after some amount of lag time for the blowdown flow to arrive at the detector

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Radiation Monitoring Lesson Plan	NOS05RMS000-17			17

L.O. Number
ABSG01E001

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 15

Given the following conditions:

- Unit 2 has experienced a tube rupture on 22 SG.
- After the reactor was tripped, the Main Turbine failed to automatically trip, and was manually tripped using the Turbine Trip handle on the control console.
- All feedwater flow was isolated to 22 SG during performance of EOP-TRIP-1, Reactor Trip or Safety Injection.
- 21, 23, and 24 SG NR levels are off-scale low.
- 22 SG NR level is 1% and rising slowly.
- Operators are currently performing actions in EOP-SGTR-1, Steam Generator Tube Rupture.
- Prior to performing the RCS cooldown to Target Temperature, operators are assessing AFW flow status.

Which of the following describes how AFW flow to 22 SG should be controlled?

a. Leave AFW flow isolated to 22 SG for the remainder of the event regardless of 22 SG NR level.

b. Establish AFW flow to 22 SG until its NR level is > 9%, then isolate AFW flow to 22 SG. Maintain 22 SG NR level >9%.

c. Establish AFW flow to 22 SG until its NR level is >19%, then isolate AFW flow to 22 SG. Maintain 22 SG NR level >19%.

d. Leave AFW flow isolated to 22 SG until the cooldown to Target Temperature is complete, then establish AFW flow as required to keep the SG tubes covered.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000038A144 EA1.44 **RO Value:** 3.4* **SRO Value:** 3.4 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Steam Generator Tube Rupture 038

KA Statement: Ability to operate and / or monitor the following as they apply to Steam Generator Tube Rupture:
Level operating limits for S/Gs

Explanation of Answers: 55.41.b(4,10) Step 6 of SGTR-1 directs AFW flow be established to the ruptured SG if its NR level is <9%. A is incorrect but plausible if it is thought that you never feed a ruptured SG if intact SG's are available. C is incorrect but plausible since it is the correct action, but the incorrect level. Note: 19% is the new (REV. 30 of EOPs) SG NR level above which intact SGs are directed to be maintained at. D is incorrect but plausible if it is thought that the purpose of leaving ruptured SG isolated during cooldown is to minimize spread of contamination IF intact SGs are available for cooldown (which is asked for in step 10.1 for initiating CD to Target Temp).

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Steam Generator Tube rupture	2-EOP-SGTR-1F	Sheet 2		30

L.O. Number
SGTR01E006

Objectives

Material Required for Examination	

Question Source:	New	Question Modification Method:		Used During Training Program	☐

Question Source Comments	

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 16

Given the following conditions:

- Unit 2 has experienced a rupture of the 24 main steamline in the mechanical penetration area upstream of 24MS167.
- The Rx is tripped and a MSLI performed.
- Feed flow is isolated to 24 SG.
- Operators are preparing to transition out of EOP-TRIP-1.

Which of the following describes how the RCS cooldown **rate** will be affected over the next 30 minutes?

The RCS cooldown rate will...

a. remain constant until 24 SG blows dry.

b. lower as the faulted SG pressure lowers.

c. remain constant until AFW flow is re-initiated to 24 SG, at which time it will increase.

d. lower until at least one intact SG NR level rises above 9%, then remain constant as AFW flow is lowered.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000040K102 AK1.02 **RO Value:** 3.2 **SRO Value:** 3.6 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Steam Line Rupture 040

KA Statement: Knowledge of the operational implications of the following concepts as they apply to Steam Line Rupture:
Leak rate versus pressure change

Explanation of Answers: 55.41.b(4,5,14) As the faulted SG pressure lowers due to the steam break, break flow will lower, and the rate at which the RCS is being cooled due to break will lower as a result. A is incorrect but plausible if it is thought that a static break will pass the same lbm of steam flow during the entire event. C is incorrect but plausible if it is thought that feed flow is initiated to keep the SG tubes wet, which only would be performed in LOSC-2 if ALL SGs were faulted. D is incorrect but plausible if it is thought that the lowering of AFW flow, which would put LESS cold FW flow into SG, would offset the lowering steam flow effect on RCS cooldown.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision

L.O. Number | Objectives

MSTEAME016

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 17

Given the following conditions:

- Unit 2 is operating at 100% power, MOL.
- 21 SGFP trips.
- NO operator action is taken in response to the SGFP trip, and the Rx does NOT trip.

Which of the following is an **UNEXPECTED** alarm if it is locked in 2 minutes after 21 SGFP trips, and what procedure would be used to address the condition associated with that unexpected alarm?

a. OHA G-3, EHC SYS TRBL. S2.OP-AB.TRB-0001, Turbine Trip Below P-9.

b. Console Alarm RC PRESS DEVIATION HI. S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction.

c. OHA G-44, COND POL TRBL. S2.OP-AB.CN-0001 Main Feedwater/Condensate System Abnormality.

d. Console Alarm RC LOOPS TAVG-TREF DEVIATION. S2.OP-AB.ROD-0001, Immovable/Misaligned Control Rods.

Answer	b	Exam Level	R	Cognitive Level	Application	Facility:	Salem 1 & 2	ExamDate:	12/19/2016

KA:	000054G445	2.4.45	RO Value:	4.1	SRO Value:	4.3	Section:	EPE	RO Group:	1	SRO Group:	1	55.43

System/Evolution Title Loss of Main Feedwater 054

KA Statement: Ability to prioritize and interpret the significance of each annunciator or alarm.

Explanation of Answers: 55.41.b(7,10) The condensate polisher trouble alarm will be in alarm due to the CN108s AND the CN109 being open at the same time. D is incorrect because RC loops Tavg-Tref deviation will be expected as rods are driving in due to the turbine runback to 65%. The ARP has operators place rods in manual, and if not successful at restoring conditions, going to AB.ROD-01. B is correct because the RC pressure deviation would not be expected, since the setpoint (+75 psig deviation)equates to when the spray valves are full open. The spray valves should be shut after the insurge due to the load rejection and then the large amount of inward rod motion. The alarm response directs entry into AB.PZR-0001. A is incorrect because G-3 will be in alarm since it receives input from the EHC Control and Status computer, which has a Loss of Feed pump Runback alarm in, and the ARP directs performance of SO.TRB-2 to reduce power if a steam valve has shut. This question meets the K/A because the operator must be able to distinguish between expected and unexpected alarms for the loss of feedwater, and prioritize them in order to know which would require action in an abnormal Response Procedure.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Main Condensate Feedwater Abnormality	S2.OP-AB.CN-0001			28

L.O. Number
CN&FDWE016

Objectives

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments	113267				

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 18

Given the following conditions:

- Unit 2 was operating at 100% power when a loss of all AC power occurred.
- 15 minutes after the power loss, operators have locally started 2B EDG.

Which of the following is an action that is REQUIRED to have been performed PRIOR to energizing 2B 4KV Vital bus, and why?

a. Shed non-essential DC loads to extend the time the Vital Instrument Inverters can power their AC loads.

b. Initiate and reset SI to prevent the auto start of a centrifugal charging pump and possible thermal shock to the RCP seals.

c. Deenergize ALL SECs and depress stop PBs for SEC actuated components to prevent overloading the 2B 4KV vital bus.

d. Start the Station Blackout Compressor to provide air for operation of 21-24AF11, AUX FEED-S/G LEVEL CONTROL VLVS, to prevent over feeding the SGs when 22 AFW pp starts.

Answer	c	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	000055A203	EA2.03	**RO Value:**	3.9	**SRO Value:**	4.7	**Section:**	EPE	**RO Group:**	1	**SRO Group:**	1	55.43

System/Evolution Title Station Blackout | 055

KA Statement: Ability to determine and interpret the following as they apply to Station Blackout:
Actions necessary to restore power

Explanation of Answers: 55.41.b(10) The Continuous Action Step for energizing a deenergized vital bus with an EDG comes AFTER the step to deenergize all SEC's. The Bases Document states on page 15 that the reason to deenergize the SECs and depress the Stop PB for all SEC controlled safety related loads is to prevent the bus from overloading. It additionally states that a further reason is to prevent charging pump automatic start and possible thermal shock to the RCP seals. SI is initiated at Step 21 NOT to prevent a charging pump from running, but rather to prevent the SI actuated valve realignment that will occur if an SI signal is sensed after power is restored. Non essential DC loads are shed at Step 35 to extend the batteries power capability. The SBO is started as part of Blackout Coping Actions in Attachment 2 Part A of AB.LOOP-1. All the distracters are actions which will be taken during an extended loss of all AC power, but the correct answer is the only one that is required to be performed AND has the correct reason for doing it prior to power restoration. D will be performed, but it is NOT the correct reason, and is required within 60 minutes of Blackout. A and B will be performed, but are not required to be performed prior to power restoration.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of All AC Power	2-EOP-LOPA-1		17	30

L.O. Number
LOPA00E007

Objectives

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 19

Given the following conditions:

- Unit 1 is operating at 100% power.
- The crew is performing S1.OP-ST.SJ-0001, Inservice Testing - 11 Safety Injection Pump.
- 11 SI pump is running.
- A loss of all off-site power occurs.
- 1B 4KV vital bus locks out on Bus Differential.

Which of the following describes Safety Injection pump status 2 minutes after the loss of off-site power?

a. ONLY 11 SI pump is running.

b. ONLY 12 SI pump is running.

c. BOTH SI pumps are running.

d. NEITHER SI pump is running.

Answer	d	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	000056A203	AA2.03	**RO Value:**	3.8	**SRO Value:**	3.9	**Section:**	EPE	**RO Group:**	1	**SRO Group:**	1	55.43

System/Evolution Title Loss of Off-Site Power | 056

KA Statement: Ability to determine and interpret the following as they apply to Loss of Off-Site Power:
Operational status of safety injection pump

Explanation of Answers: 55.51.b(8) neither SI pump will be running after a loss of off-site power with NO SI required. A is incorrect but plausible if it is thought that the 11 SI pump breaker would remain shut during the response, and the pump would restart. B is incorrect but plausible if it is thought that only 12 SI pump would start because 11 SI pump was originally is an abnormal configuration. C is incorrect but plausible if it is thought the SECs would load both SI pumps, which it would only do if there was an accident signal coupled with the Blackout signal. D is correct because the SEC's will strip the 4KV vital bus breakers before sequencing on BLACKOUT loads, and neither SI pump would start. 11/12 SI pumps are powered from A / C 4KV vital buses.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Offsite Power	S1.OP-AB.LOOP-0001	Attachment 3	58	31

L.O. Number
ABLOP1E001

Objectives

Material Required for Examination

Question Source:	New	**Question Modification Method:**		**Used During Training Program**

Question Source Comments

Comment

RO SkyScraper	SRO Skyscraper	RO System/Evolution List	SRO System/Evolution List	Outline Changes

Question Topic RO 20

Given the following conditions:

- Unit 2 is at 100% power
- A loss of 2A 115 VAC Vital Instrument Bus occurs.

Which of the following describes the impact of the loss on the AFW system if a reactor trip **with no SI** occurs before the bus is recovered?

a. 21 AFW Pump will have to be started manually.

b. Operators must dispatch an NEO to locally throttle 23AF21 and 24AF21.

c. 21 AFW Pump may trip during the automatic start because 23AF21 and 24AF21 are failed open.

d. Operators must press the PRESSURE OVERRIDE DEFEAT pushbutton for 21 AFW Pump before gaining control of 23AF21 and 24AF21.

Answer	b	Exam Level	R	Cognitive Level	Application	Facility:	Salem 1 & 2	ExamDate:	12/19/2016

KA:	000057A106	AA1.06	RO Value:	3.5	SRO Value:	35	Section:	EPE	RO Group:	1	SRO Group:	1	55.43

System/Evolution Title Loss of Vital AC Instrument Bus 057

KA Statement: Ability to operate and / or monitor the following as they apply to Loss of Vital AC Instrument Bus:
Manual control of components for which automatic control is lost

Explanation of Answers: 55.41.b(7) B is correct because 23AF21 and 24AF21 fail closed due pressure override as power is lost to the pressure transmitter. An operator is dispatched at Step 13 of AB. Power is also lost to the valve controllers. A is incorrect because 21 AFW Pump would only have to be started if an SEC actuation occurred, and the stem states no SI has occurred. C is incorrect because the valves fail closed, D is incorrect because power is not available to the valve controllers.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of 2A 115 VAC Vital Instrument Bus	S2.OP-AB.115-0001		4	20

L.O. Number
AB1151E003

Objectives

Material Required for Examination

Question Source:	Facility Exam Bank	Question Modification Method:	Direct From Source	Used During Training Program	

Question Source Comments 80310

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 21

Given the following condition:

- Operators are performing actions in 2-EOP-FRCC-1 , Response to Inadequate Core Cooling.

Which of the following identifies how the PZR PORVs will be operated after starting ALL available RCPs with CET temperatures above 1200°F?

a. Shut BOTH PORVs.

b. Shut ONLY ONE PORV.

c. Open ONLY ONE PORV.

d. Open BOTH PZR PORVs.

Answer	d	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	000074K202	EK2.02	**RO Value:**	3.9	**SRO Value:**	4.0	**Section:**	EPE	**RO Group:**	2	**SRO Group:**	2	55.43

System/Evolution Title Inadequate Core Cooling 074

KA Statement: Knowledge of the interrelations between Inadequate Core Cooling and the following:
PORV

Explanation of Answers: 55.41.b(10) RCPs will be started at step 25 if CETs are still >1200°F, to clear the water in the RCS intermediate leg and permit the circulation of hot gases from the overheated core to circulate through the steam generators. If RCP(s) restart is not effective in decreasing CET temperature <1200°F (it is not per stem), the PZR PORVs (BOTH) will be opened at step 25.3 to help reduce RCS pressure to allow ECCS injection.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Response to Inadequate Core Cooling	2-EOP-FRCC-1	Basis Doc	38	30

L.O. Number | Objectives

FRCC00E002

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 22

Given the following conditions:

- Unit 2 has experienced a SBLOCA.
- Operators have transitioned from 2-EOP-TRIP-1 Reactor Trip Response, to 2-EOP-LOCA-6 LOCA Outside Containment.
- The 21SJ49 COLD LEG ISOLATION VALVE has just been closed in an attempt to isolate the leak.

Which choice describes the response which would indicate successful leak isolation, and the next action to be performed IAW 2-EOP-LOCA-6?

a. RCS pressure stable, open 21SJ49.

b. RCS pressure stable, stop 22 RHR pump.

c. RCS pressure rising, stop 21 RHR pump.

d. RCS pressure rising, close 21RH19 RHR DISCHARGE X-CONN.

Answer c **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE04A101 EA1.1 **RO Value:** 4.0 **SRO Value:** 4.0 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title LOCA Outside Containment E04

KA Statement: Ability to operate and / or monitor the following as they apply to LOCA Outside Containment:
Components, and functions of control and safety systems, including instrumentation, signals, interlocks, failure modes, and automatic and manual features.

Explanation of Answers: 55.41.b(10) Distractor a is incorrect because after the leak is isolated by closing the 21SJ49, the procedure leaves it closed. Also stable RCS pressure is not the required RCS pressure indicating leak is isolated. C is correct because rising RCS pressure indicates leak isolation, and the next step is to stop 21 RHR pump because the RHR system is split and there is no discharge path. Distractor D is incorrect because the 21RH19 would already have been closed at step 2. Distractor b is incorrect because both stable pressure and stopping the opposite loop RHR pump are not correct.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
LOCA Outside Containment	2-EOP-LOCA-6			30

L.O. Number

LOCA06E002

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 127086

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: RO 23

Given the following conditions:

- A loss of heat sink has occurred.
- The operating crew is establishing RCS Bleed and Feed in accordance with EOP-FRHS-1, Loss Of Secondary Heat Sink.
- The RO opens one PORV. He reports that the second PORV will NOT open.

Which one of the following describes the consequences of the PORV failure?

a. ALL SGs will require depressurization to inject the alternate source of feedwater.

b. The RCS will rapidly re-pressurize when the SGs empty, resulting in a violation of a RCS Safety Limit.

c. Bleed and Feed cooling of the RCS must be terminated and secondary depressurization to inject condensate pump flow must be immediately initiated.

d. The RCS may not depressurize quickly enough to ensure sufficient SI flow to provide RCS heat removal, and other RCS openings may have to be established.

Answer: d **Exam Level:** R **Cognitive Level:** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE05K301 EK3.1 **RO Value:** 3.4 **SRO Value:** 3.8 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title: Loss of Secondary Heat Sink E05

KA Statement: Knowledge of the reasons for the following responses as they apply to Loss of Secondary Heat Sink:
Facility operating characteristics during transient conditions, including coolant chemistry and the effects of temperature, pressure, and reactivity changes and operating limitations and reasons for these operating characteristics.

Explanation of Answers: 55.41.b(8,10) Distractor c is incorrect because Bleed and Feed is not terminated if only one PORV is open. Distractor b is incorrect because the RCS Safety limit is 2735 psig and will not occur with the PORV open. Distractor a is incorrect because action to align condensate pumps is already taken, and not as a contingency to Bleed and Feed. D is correct because FRHS Basis document describes the consequences of not having both PORV's open, and it is D.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Response to Loss of Secondary Heat Sink	2-EOP-FRHS-1		32	31

L.O. Number: FRHS00E009

Objectives

Material Required for Examination:

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program:** ☐

Question Source Comments:

Comment:

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 24

Given the following conditions:

- Unit 2 has experienced a steam line break inside containment.
- Operators have entered FRTS-1, Response to Imminent Pressurized Thermal Shock.

Why will the operators be instructed to terminate SI and start RCP(s) if possible?

a. The soak required by FRTS-1 requires SI to be secured. RCPs are started to provide the ability to use spray to depressurize the primary.

b. The soak required by FRTS-1 requires SI to be secured. RCPs are started to provide mixing of cold SI and warm reactor coolant water.

c. Safety Injection flow is a significant contributor to any cold leg temperature decrease or overpressure condition and must be terminated. RCPs are started to minimize temperature gradient across S/G tube sheets.

d. Safety Injection flow is a significant contributor to any cold leg temperature decrease or overpressure condition and must be terminated. RCPs are started to provide mixing of cold SI and warm reactor coolant water.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE08K101 EK1.1 **RO Value:** 3.5 **SRO Value:** 3.8 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Pressurized Thermal Shock E08

KA Statement: Knowledge of the operational implications of the following concepts as they apply to Pressurized Thermal Shock:
Components, capacity, and function of emergency systems.

Explanation of Answers: 55.41.b.(10,8)A- incorrect - purpose for RCPs is not priority in FRTS-1, soak is not basis for SI. B - incorrect - soak not basis. C - incorrect - SI basis correct, RCP basis not accurate. D-correct -page 12-13

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Response to Imminent Pressurized Thermal Sh	2-EOP-FRTS-1			30

L.O. Number | **Objectives**

FRTS00E007

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 73425.

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 25

Given the following conditions:

- Unit 2 has tripped from 100% power due to a Loss of Off-Site Power.
- Operators are performing a cooldown IAW 2-EOP-TRIP-6 NATURAL CIRCULATION RAPID COOLDOWN WITH RVLIS.

Which choice identifies the MINIMUM RVLIS Full Range level required to be maintained during the cooldown, and its significance?

a. 70% to limit steam entering the RCS hot legs.

b. 100% to limit steam entering the RCS hot legs.

c. 70% to ensure positive level indication of RCS.

d. 100% to ensure positive level indication of RCS.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE10K303 EK3.3 **RO Value:** 3.4 **SRO Value:** 3.6 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Natural Circulation with Steam Void in Vessel with/without RVLIS E10

KA Statement: Knowledge of the reasons for the following responses as they apply to Natural Circulation with Steam Void in Vessel with/without RVLIS:
Manipulation of controls required to obtain desired operating results during abnormal, and emergency situations.

Explanation of Answers: 55.41.b(10)74% is minimum allowed at step 10, and get into a do loop until it is satisfied. The Bases Document states that if steam enters the hot legs, there may be some potential for it to reach the top of the SG U tubes, thereby disrupting the natural circulation flow circuit. By monitoring RVLIS and limiting the void growth to the top of the hot legs, the potential for introducing voids into the SG Utubes is minimized. Setpoint was changed from 74% to 70% current revision of EOPs.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Natural Circulation Rapid Cooldown with RVLIS	2-EOP-TRIP-6	Bases Document	22	30

L.O. Number
TRP004E004

Objectives

Material Required for Examination

Question Source: Previous 2 NRC Exams **Question Modification Method:** Editorially Modified **Used During Training Program**

Question Source Comments Used on Salem December 2014 NRC exam (2 exams ago.) Setpoint was changed from 74% to 70% current revision of EOPs in Dec 2015).

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 26

Given the following conditions:

- A LBLOCA occurred on Unit 2.
- 2C 4KV vital bus locked out on Bus Differential.
- Both 22 and 24 CFCUs failed to start in low speed and cannot be started.
- After entering 2-EOP-LOCA-3, Transfer to Cold Leg Recirculation when required, the crew has transitioned to EOP-LOCA-5, Loss of Emergency Recirculation when neither SJ44 valve opened.
- Containment pressure is 18 psig.
- The crew is performing Step 9 to determine the required number of Containment Spray pumps to be operated.

Which of the following identifies the Containment Spray pump(s) which will be in service if Step 9 is completed seven minutes after the RWST Lo Level alarm was received?

a. BOTH Containment Spray pumps.

b. NEITHER Containment Spray pump.

c. 21 Containment Spray pump ONLY.

d. 22 Containment Spray pump ONLY.

Answer c **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE11K102 EK1.2 **RO Value:** 3.6 **SRO Value:** 4.1 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Loss of Emergency Coolant Recirculation E11

KA Statement: Knowledge of the operational implications of the following concepts as they apply to Loss of Emergency Coolant Recirculation: Normal, abnormal and emergency operating procedures associated with (Loss of Emergency Coolant Recirculation).

Explanation of Answers: 55.41.b(10) Table C uses RWST level, Containment pressure, and the number of operating CFCUs in Low speed to determine how many containmernt spray pumps are required to be run. LOCA-3 was entered with RWST level at 15.2', and with 2C 4KV vital bus deenergized will be depleting at ~13,000 gpm, with 2 RHR, 1 charging, 1 SI, and 1 CS pump running. This will require over 9 more minutes of injection before RWST level would be lower than 1'. With the stem stating that Step 9 is complete 7 minutes after 15.2' in RWST, then Table C will indicate that ONE CS pump will be required. 22 CS pump does not have power with C vital bus deenergized, so 21 will be run. If RWST drain down calculation is incorrectly performed and it is assumed RWST level is <1', then NO CS pumps would be required. The BOTH distracter is for RWST level between 1-15.2' if either containment presure is >47 psig, or if no CFCUs were running.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Emergency Recirculation	2-EOP-LOCA-5	Basis Doc	28	30

L.O. Number
LOCA05E006

Objectives

Material Required for Examination	RO 26 2-EOP-LOCA-5 Sheet 1 AND S2.OP-TM.ZZ-0002 TANKS CURVE RWST

Question Source:	New	**Question Modification Method:**		**Used During Training Program**	☐

Question Source Comments	

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 27

Following a SBLOCA inside containment concurrent with a faulted SG inside containment, the following conditions are noted:

- RCS subcooling is indicating 10°F.
- RCS pressure is 1380 psig.
- Containment spray actuation has occurred, but neither Containment Spray pump started.
- The crew is implementing 2-EOP-FRCE-1, Response to High Containment Pressure.

Which of the following describes why the Reactor Coolant Pumps (RCP) are tripped under these conditions?

a. To prevent RCP motor damage due to lack of cooling.

b. To minimize additional loss of reactor coolant from RCP seal leakoff.

c. To prevent subsequent thermal barrier damage due to backflow of RCS fluid.

d. To minimize possible subsequent core damage due to pumping a two-phase mixture.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE14K102 EK1.2 **RO Value:** 3.2 **SRO Value:** 3.7 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title High Containment Pressure E14

KA Statement: Knowledge of the operational implications of the following concepts as they apply to High Containment Pressure: Normal, abnormal and emergency operating procedures associated with (High Containment Pressure).

Explanation of Answers: 55.41.b(10,3) CS actuation occurs at 15 psig in containment, as well as Phase B isolation, which shuts RCP CCW cooling inlet and outlet valves. Per FRCE-1 basis document page 4, "...RCPs are tripped since component cooling water to the RCP seals and motors is isolated." B is incorrect but plausible if RCP seal package operation is not understood (running pump to secured pump). C is incorrect but plausible based on having no CCW flow to thermal barrier. D is incorrect but plausible if it is thought a two phase mixture could be present (subcooling is indicated in stem).

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Response to Excessive Containment Pressure	2-EOP-FRCE-1			30

L.O. Number

FRCE00E006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program** ☐

Question Source Comments Replaced a distracter which used to have containment spray water in it, as new rev of EOPs requires CS pumps NOT running to enter FRCE-1.

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 28

Given the following conditions:

- Unit 2 is operating at 4% power during a plant startup.
- 22 RCP trips.

Which of the following identifies the **Tech Spec required** action for the RCP trip?

a. Be in Mode 3 within 1 hour.

b. Be in Mode 3 within 15 minutes.

c. Immediately initiate corrective action to return 22 RC loop to operable status as soon as possible.

d. Suspend all operations involving a reduction in boron concentration of the reactor coolant system.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 003000G222 2.2.22 **RO Value:** 4.0 **SRO Value:** 4.7 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Reactor Coolant Pump System 003

KA Statement: Knowledge of limiting conditions for operations and safety limits.

Explanation of Answers: 55.41.b(3,10) With the reactor in Mode 2, LCO 3.4.1.1 states all reactor coolant loops shall be in operation. The ACTION with less than all loops in operation is to be in HSB within 1 hour. B is incorrect but plausible as it has the correct action but wrong time. C and D are incorrect but plausible as they are actions taken for Mode 4 and Mode 3 RCP operation respectively.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs		Section 3/3 4.1	3/4 4-1	282

L.O. Number TECHSPE015

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper	SRO Skyscraper	RO System/Evolution List	SRO System/Evolution List	Outline Changes

Question Topic RO 29

Given the following conditions:

- Unit 1 is operating a 100% power.
- After performing a normal dilution for RCS temperature control, the START pushbutton for CVCS Makeup Mode Select was NOT depressed when returning the CVCS Makeup System to automatic.

Which of the following describes the effect this will have if VCT level were to continuously lower with NO operator action?

a. 1SJ1 and 1SJ2, RWST to Charging Pump Suction valves will open automatically when VCT level reaches the lo-lo level setpoint, and 1CV40 and 1CV41, VCT Outlet Isolation Valves, will shut automatically.

b. 1SJ1 and 1SJ2, RWST to Charging Pump Suction valves will open automatically when VCT level reaches the lo-lo level setpoint, and 1CV40 and 1CV41, VCT Outlet Isolation Valves will remain open.

c. A dilution will occur when the Primary Water pumps start at 14% VCT level with no boration flow aligned to the CVCS Blender.

d. The VCT will empty and the charging pumps will lose suction.

Answer a **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 004000K404 K4.04 **RO Value:** 3.2 **SRO Value:** 3.1 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Chemical and Volume Control System 004

KA Statement: Knowledge of Chemical and Volume Control System design feature(s) and or interlock(s) which provide for the following: Manual/automatic transfers of control

Explanation of Answers: 55.41.b(7) The CVCS START PB must be depressed to align the CVCS M/U system to whichever mode was selected (BORATE, DILUTE, ALT DILUTE, AUTO). D is incorrect but plausible if it is thought that the SJ1 auto open interlock is based on having the CVCS M/U Mode Selector in AUTO and START. C is incorrect but plausible if it is thought that Primary Water pump operation would occur based on last position of CVCS M/U system (dilute per stem) when the auto makeup setpoint would be reached at 14% lowering VCT level. The SJ1 and SJ2 open automatically on 2/2 level inst lo-lo VCT level, and once they are open the CV40 and CV41 shut to transfer suction of charging pumps to RWST.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
1SJ1 Schematic	211577			3
Chemical and Volume Control Lesson Plan	NOS05VVCS00-16		43	16

L.O. Number

CVCS00E006

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 30

Given the following conditions:

- Unit 2 is operating at 100% power, steady state.
- The Charging System Master Flow Controller is in MANUAL.
- A malfunction in the H2 pressure regulator supplying cover gas to the VCT occurs, and VCT pressure rises from 25 to 35 psig.

Which of the following describes the effect this will have on CVCS when steady state conditions are established?

a. Charging flow will rise.

b. Letdown flow will rise.

c. Charging flow will lower.

d. Letdown flow will lower.

Answer	a	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	004000K526	K5.26	**RO Value:**	3.1	**SRO Value:**	3.2	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	☐

System/Evolution Title Chemical and Volume Control System 004

KA Statement: Knowledge of the operational implications of the following concepts as they apply to the Chemical and Volume Control System: Relationship between VCT pressure and NPSH for charging pumps

Explanation of Answers: 55.41.b(5,7) Raising VCT pressure will cause NPSH available to the charging pumps to rise. With the flow controller in manual, and more suction head available, pump flow will rise. The letdown pressure control valve will respond in automatic and letdown flow will remain the same. Static pressure at the pump suction: An increase in surge tank level, increase in the pressure above the liquid in the surge tank, increase in booster pump output, or opening of the pump suction valve would increase suction pressure

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
PWR Generic Fundamentals-Components			44	

L.O. Number | **Objectives**

CVCS00E004

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program** ☐

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 31

Given the following conditions:

- Unit 2 is in MODE 4.
- 21 RHR loop is in service in SDC mode.
- 22 RHR loop is O/S and aligned for ECCS.

Which of the following would cause the temperature of the water flowing to the RCS from the RHR system to RISE?

a. 21SJ49 RHR DISCH TO COLD LEGS is shut.

b. 22CC16 RHR HX COMP CLG OUT valve is shut.

c. 2RH20 RHR HX Bypass Valve is throttled open.

d. 21RH18 RHR HX FLOW CONT VALVE is throttled open.

Answer	c	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	005000A102	A1.02	**RO Value:**	3.3	**SRO Value:**	3.4	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43

System/Evolution Title Residual Heat Removal System 005

KA Statement: Ability to predict and/or monitor changes in parameters associated with operating the Residual Heat Removal System controls including:
RHR flow rate

Explanation of Answers: 55.41.b(5) A is incorrect because with the RH19s open, the discharge from 21 RHR HX will flow through the normally open 22SJ49. B is incorrect because the valve is normally shut, and with the 22RH12 shut there won't be any flow through 22 CCHX from 21 loop anyways. C is correct because bypassing flow around the in service HX will cause system temp to rise, as more warm RCS water is mixed in with the cooled RHR HX outlet water and returned to the RCS. D is incorrect because opening the 21RH18 will increase flow through the in service RHR HX, causing outlet temp to lower.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
RHR Simplified	205332-SIMP			2
Initiating RHR	S2.OP-SO.RHR-0001			29

L.O. Number RHR000E004

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 13-01 C RO 31

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: RO 32

Given the following conditions:

- Unit 1 was operating at 100% power when a SBLOCA occurred.
- During the performance of EOP-TRIP-1, RCS pressure has lowered to 1600 psig.
- 11 SI pump failed to start.
- Current TOTAL ECCS injection flow is 400 gpm.

Which of the following approximates TOTAL ECCS injection flow if 11 SI pump is successfully started?

a. 400 gpm.

b. 500 gpm.

c. 800 gpm.

d. 1000 gpm

Answer	a	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	006000A117	A1.17	**RO Value:**	4.2	**SRO Value:**	4.3	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43

System/Evolution Title: Emergency Core Cooling System | 006

KA Statement: Ability to predict and/or monitor changes in parameters associated with operating the Emergency Core Cooling System controls including:
ECCS flow rate

Explanation of Answers: 55.41.b(8) Shutoff head for an SI pump is 1520 psid. Starting one at 1600 psig in RCS (with suction from 41' of RWST level,) would provide no additional ECCS flow. The 500 gpm distracter if it is thought that shutoff head is slightly higher than 1600 psig. The 1000 gpm distracter is if ~full SI pump flow were added to ECCS flow. The 800 gpm is flow at 1082 psid.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
ECCS Lesson Plan	NOS05ECCS00		27	9

L.O. Number
ECCS00E008

Objectives

Material Required for Examination:

Question Source: Facility Exam Bank | **Question Modification Method:** Concept Used | **Used During Training Program**

Question Source Comments: 61678

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 33

Given the following conditions:

- Salem Unit 2 has experienced a LBLOCA.
- 21SJ44 fails to open when demanded and cannot be opened.

Upon completion of EOP-LOCA-3, Transfer to Cold Leg Recirculation, which of the following identifies how the 21SJ44 failure affected the Cold Leg Recirc alignment compared to if the 21SJ44 were open?

a. Containment spray header flow will be zero because 21CS36 is interlocked to remain shut.

b. Containment spray header flow will be zero because 21 RHR pump will have been stopped.

c. Containment sump recirc flow will be directed through 22SJ45 and 22CS36 ONLY because 21 RHR pump will have been stopped.

d. Containment sump recirc flow will be directed through the 22SJ49 and 22CS36 ONLY because 21SJ45 is interlocked to remain shut.

Answer c **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 006000K301 K3.01 **RO Value:** 4.1 **SRO Value:** 4.2 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Emergency Core Cooling System 006

KA Statement: Knowledge of the effect that a loss or malfunction of the Emergency Core Cooling System will have on the following: RCS

Explanation of Answers: 55.41.b(8,7) When 21SJ44 fails to open, the procedure will direct stopping 21 RHR pump, and it will remain stopped as long as the 21SJ44 is shut (step 5.2) The 21SJ45 is interlocked with the 21SJ44 such that it cannot be opened with the 21SJ44 shut. The 21CS36 interlock is with the RH1 and RH2 needing to be SHUT before the CS36 can be OPENED. A is incorrect because CS header flow will be provided by opening the 22CS36 since 22 RHR pump is running at step 22. B is incorrect because even though 21 RHR pump WILL have been stopped, the running RHR pump will be supplying CS header flow. D is incorrect because under ALL circumstances if 22 RHR pump is running when the RWST lo-lo level alarm is received, the 22SJ49 isolation to cold legs is shut. The final lineup will be with ALL containment sump recirculation flow going through the 22SJ45 to Charging/SI pump suctions, and through 22CS36 to spray headers.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
ECCS Lesson Plan			47,52	
Transfer to Cold Leg Recirculation	2-EOP-LOCA-3			30
ECCS Simplified Drawing	205350-SIMP			5

L.O. Number

ECCS00E016

ECCS00E006

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 34

Which of the following describes the operation of the PRT Vent Vent, 2PR15?

The 2PR15 recieves a(n)......

a. shut signal at 10 psig in PRT.

b. open signal at 10 psig in PRT.

c. shut signal at 100 psig in PRT.

d. open signal at 100 psig in PRT.

Answer	a	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	007000A404	A4.04	**RO Value:**	2.6*	**SRO Value:**	2.6*	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	

System/Evolution Title Pressurizer Relief Tank/Quench Tank System 007

KA Statement: Ability to manually operate and/or monitor in the control room:
PZR vent valve

Explanation of Answers: 55.41.b(7) PRT vent valve 2PR15 is interlocked to SHUT on a high PRT pressure of 10 psig. The PRT Rupture Diaphram will actuate at 100 psig in the PRT.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Control Console CC2	S2.OP-AR.ZZ-0012		51-52	39

L.O. Number
PZRPRTE006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Concept Used **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 35

Given the following conditions:

- Unit 2 is operating normally at 65%, steady state.
- 22 and 23 CCW pumps are in service.

Subsequently 22 CCW pump trips.

Which of the following describes how the CCW system will respond or be manipulated by the control room crew?

a. 21 CCW pump will auto start when CCW header pressure lowers to 70 psig.

b. 21 CCW pump will auto start immediately since less than 2 CCW pump 4KV breakers are shut.

c. The NCO will start 21 CCW pump immediately since the 3rd CCW pump is normally aligned in MANUAL to prevent 3 CCW pumps from operating simultaneously.

d. The control room crew will enter S2.OP-AB.CC-0001, Component Cooling Abnormality, isolate the non-safeguards header, and start 21 CCW pump if CCW header pressure lowers to 80 psig.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 008000K401 K4.01 **RO Value:** 3.1 **SRO Value:** 3.3 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Component Cooling Water System 008

KA Statement: Knowledge of Component Cooling Water System design feature(s) and or interlock(s) which provide for the following:
Automatic start of standby pump

Explanation of Answers: 55.41.b(7) Normal CCW system alignment is 2 CCW pumps running in MANUAL, and one CCW pump in AUTO and stopped. A CCW pump selected to AUTO will start when either 21 or 22 CCW header pressure lowers to 70 psig. There is no interlock based on breaker positions, but other system DO have a breaker anticipatory function. AB.CCW could be entered, but no actions will be performed other than ensuring the backup pump started.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Control Console CC1	S2.OP-AR.ZZ-0011		92	60

L.O. Number
CCW000E006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 111918

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 36

Given the following conditions:

- Unit 2 is operating at 4% power holding during a startup.
- 2PS1, PZR Spray Valve, fails open and cannot be shut.

With NO operator action, which of the following protective actions will occur FIRST?

a. Reactor trip on OT/DT.

b. Reactor trip on High PZR level.

c. Reactor trip on Low PZR pressure.

d. Safety Injection on Low PZR pressure.

Answer	d	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	010000A202	A2.02	**RO Value:**	3.9	**SRO Value:**	3.9	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	☐

System/Evolution Title Pressurizer Pressure Control System 010

KA Statement: Ability to (a) predict the impacts of the following on the Pressurizer Pressure Control System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:

Spray valve failures

Explanation of Answers: 55.41.b(7)Low pressure Rx trip is not reinstated until > P-10. D/T is very small at 4% power and would not initiate any protective action, but it would be the first protective action to occur at 100% power. PZR level will not rise to the Rx trip setpoint. The low pressure SI was reinstated during heatup/pressurization when RCS pressure was >1915 psig.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Fluency List	NOS05FLUNCY-09			9

L.O. Number
FLUNCYE002

Objectives

Material Required for Examination

Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐

Question Source Comments 122719

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 37

With normal Letdown in service, all of the following conditions would cause any open Letdown Orifice Isolation Valve 2CV3, 2CV4, or 2CV5 to automatically shut EXCEPT:

a. Phase A isolation signal occurs.

b. PZR level lowers to less than 17%.

c. 2CV2 OR 2CV277, Letdown Isolation Valve is shut from 2CC2.

d. 2CV7, Letdown Line Containment Isolation Valve is shut from 2CC2.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 011000A405 A4.05 **RO Value:** 3.2 **SRO Value:** 2.9 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Pressurizer Level Control System 011

KA Statement: Ability to manually operate and/or monitor in the control room:
Letdown flow controller

Explanation of Answers: 55.41.b(7) The 2CV7 is NOT interlocked with the Orifice Isolation Valves. All the other conditions will cause letdown flow to be isolated by closing the CV3,4,5.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Logic Diagram	224431			5

L.O. Number
CVCS00E006

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 38

Which of the following actions changes the setpoint at which a Rx trip would occur while performing a power ascension during a plant startup?

a. Placing a PRNI drawer Rate Mode switch to Reset at 4% Rx power.

b. Placing the High Flux at Shutdown Block switches in Block at 4% Rx power.

c. Depressing the Block Power Range A and B pushbuttons at 12% Rx power.

d. Depressing the Reset Source Range A and B pushbuttons at 12% Rx power.

Answer c **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 012000A101 A1.01 **RO Value:** 2.9* **SRO Value:** 3.4 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Reactor Protection System 012

KA Statement: Ability to predict and/or monitor changes in parameters associated with operating the Reactor Protection System controls including: Trip setpoint adjustment

Explanation of Answers: 55.41.b(7) A is incorrect but plausible if it is thought that resetting a High Flux Rate trip on a PRNI drawer would affect the Rx power trip setpoint. C is correct because it blocks the low power Rx trip (25%) when above 10% power. B is incorrect but plausible if it were to be confused with the Source Range Block switches which blocks the Source Range hi power trip. D is correct but plausible as it WOULD change the setpoint if it were performed below 10% power after the SR hi power trip was blocked, but it is prevented from re-energizing the SRNI's (which would lower the Rx power trip setpoint to 1X10E5 cps) by the P-10 block.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Rx Protection Nuclear Instr Trip Signals	221052			7

L.O. Number **Objectives**

RXPROTE012

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 39

Given the following conditions:

- Unit 2 is operating at 100% power.
- 21 and 22 CCW pumps are in service.
- 23 Charging pump is in service.
- A Safety Injection is initiated.

When performing SEC loading verification in TRIP-1, the PO reports the following equipment is not running:

- 21 Charging pump
- 21 Safety Injection pump
- #2 Emergency Air Compressor

Assuming the SEC BLOCK switches on 2RP1 DO NOT WORK, which of the following identifies ONLY the power supplies which must be deenergized in order to start the above equipment?

a. 2AVIB24, 2BVIB27.

b. 2AVIB24, 2CVIB9.

c. 2BVIB27, 2CVIB9.

d. 2AVIB24, 2BVIB27, 2CVIB9.

Answer d **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 013000K201 K2.01 **RO Value:** 3.6* **SRO Value:** 3.8 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Engineered Safety Features Actuation System 013

KA Statement: Knowledge of bus power supplies to the following:
ESFAS/safeguards equipment control

Explanation of Answers: 55.41.b(7) The A, B, C SEC's are powered from their respective vital instrument buses (VIB), and control the equipment powered from the respective 4KV and 460 Volt buses. 21 Charging pump is powered from B vital bus. 21 SI pump is powered from A vital bus. #2 ECAC is powered from C vital bus.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Rx Trip or Safety Injection	2-EOP-TRIP-1		1	30
4KV Vital buses one line	203061			35
2C 460V vital bus one line	601392			24

L.O. Number
SEC000E011

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 40

Given the following conditions:

- Unit 2 was tripped from 100% power about 16 minutes ago.
- IRNI 2N35 indicates 2.0 E-11 Amps.
- IRNI 2N36 indicates 2.0 E-10 Amps.
- Channel I SUR is -0.3 dpm
- Channel II SUR is -0.06.

Which of the following describes the condition present, and any action(s) required to be performed as a result of this condition?

a. 2N35 is over compensated. Manually energize Source Range channels.

b. 2N36 is under compensated. Manually energize Source Range channels.

c. 2N35 is under compensated. Ensure Source Range channels automatically energize when 2N36 lowers to 7.0 E-11 Amps.

d. 2N36 is over compensated. Ensure Source Range channels automatically energize when 2N36 lowers to 7.0 E-11 Amps.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 015000A202 A2.02 **RO Value:** 3.1 **SRO Value:** 3.5* **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Nuclear Instrumentation System 015

KA Statement: Ability to (a) predict the impacts of the following on the Nuclear Instrumentation System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:

Faulty or erratic operation of detectors or compensating components

Explanation of Answers: 55.41.b(6,10) TRIP-2 step 22 asks if both IRNI channels are reading <7E-11 A. If they are not, it asks if under compensation is preventing proper operation. If yes then the operator is instructed to manually energize SR channels. IRNI channels normally continue to lower until off scale on control console. If the reading is higher that expected, and SUR is abnormally low as indicated in stem, this provides justification for an undercompensated instrument, which allows more lower energy gammas to be seen. On instrument

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Reactor Trip Response	2-EOP-TRIP-2		Sh 4	30

L.O. Number

EXCOREE009

EXCOREE010

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 158886

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 41

Given the following conditions:

- Unit 1 is operating normally at 100% power.

Which of the following identifies how CFCU's would be affected if a MODE 1 SEC initiation signal occurs?

a. ONLY the running CFCU's will swap from High to Slow speed.

b. ONLY the running CFCU's will swap from Low to High speed.

c. Running CFCU's will stop, then ALL CFCU's will start in High Speed.

d. Running CFCU's will stop, then ALL CFCU's will start in Low Speed.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 022000A301 A3.01 **RO Value:** 4.1 **SRO Value:** 4.3 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Containment Cooling System 022

KA Statement: Ability to monitor automatic operations of the Containment Cooling System including:
Initiation of safeguards mode of operation

Explanation of Answers: 55.41.b(7) By procedure, a maximum of 4 CFCUs can be running in High Speed. That would make choices a and b incorrect based on the 5th non-running CFCU which would start in Low Speed along with the 4 CFCU's which received a stop signal then a start signal. CFCU's swapping to low speed receive a stop signal, then a time delay to allow motor to coast down, before the Low speed start signal occurs. During a MODE 1 SEC initiation, (SI with vital power) ALL CFCUs running in high speed will be stopped, then ALL CFCUs will start in LOW speed. A and B are incorrect but plausible if it is thought that only the running CFCUs will swap speed. C is incorrect because speed is wrong.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Safeguards Equipment Control System	NOS05SEC000-07		14-15	7

L.O. Number

CONTMTE007

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 42

21 CFCU is running in Low speed during its weekly exercise and flush.

Which of the following identifies how steady state service water flow through 21 CFCU would be affected when the CFCU is transferred from Low Speed to High Speed?

Service water flow will...

a. rise.

b. lower.

c. remain the same.

d. rise or lower based on initial SW header pressure.

Answer	c	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	022000K101	K1.01	**RO Value:**	3.5	**SRO Value:**	3.7	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	☐

System/Evolution Title Containment Cooling System 022

KA Statement: Knowledge of the physical connections and/or cause-effect relationships between Containment Cooling System and the following: SWS/cooling system

Explanation of Answers: 55.41.b(9,8) The CFCU SW flow control valve SW223 has a position limiter on it, typically 50% travel. The SW223 opens on a start signal from either low speed or high speed. With a mechanical stop employed, SW flow will be the same for high speed or low speed operation. The stem states "steady state SW flow" because the CFCU is normally stopped for 30 seconds when transferring speed. Distracter D is plausible if it is thought that SW header pressure would change, which would affect SW system flows, but in steady state to steady state comparisons it would not

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
IST CFCU SW Valves	S2.OP-ST.SW-0010		18	20

L.O. Number
CONTMTE007

Objectives

Material Required for Examination

Question Source:	Previous 2 NRC Exams	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐

Question Source Comments Salem Dec 2014 NRC exam

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 43

Given the following conditions:

- Unit 1 has experienced a LOCA.
- Control room operators have progressed to the point where the SECs have been reset.
- Containment pressure was 4.5 psig when the SECs were reset.

Which of the following describes the Containment Spray System response should a HI-HI containment pressure signal be generated at this point in the accident?

CS system valves...

a. would realign for spray, the CS pumps would be started by the SEC.

b. would realign for spray, the CS pumps would have to be manually started.

c. must be manually realigned for spray, but the CS pumps would be started by the SEC.

d. must be manually realigned for spray, the CS pumps would have to be manually started.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 026000A301 A3.01 **RO Value:** 4.3 **SRO Value:** 4.5 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Containment Spray System 026

KA Statement: Ability to monitor automatic operations of the Containment Spray System including:
Pump starts and correct MOV positioning

Explanation of Answers: 55.41.b(7,8,9)The SEC controller operates the CS pumps at 2 different point in the sequence UNTIL the SEC is reset. The SEC ONLY control the CS pumps, not the CS valves. The valve realign on the hi-hi containment pressure signal whenever it is received, but once the SEC is reset, it will not start the CS pumps since the sequencer is no longer active.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Containment Spray Lesson Plan	NOS05CSPRAY-06		42-43	6

L.O. Number

CSPRAYE006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 80567

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 44

Given the following conditions:

- Unit 2 is performing actions in 2-EOP-LOCA-1 for a LOCA.
- Containment pressure peaked at 20 psig.

When performing CS Pump Stop Criteria steps in LOCA-1, the RO depresses both Reset Spray Actuation pushbuttons with Containment pressure at 15.1 psig.

Which of the following describes the effect of this action, if anything?

a. There will be no effect.

b. Containment Spray valves 21/22CS2, 2CS14, 2CS16, and 2CS17 will shut.

c. Containment Spray Actuation signal will reset and NOT reinitiate after the Reset Spray Actuation pushbuttons are released.

d. Containment Spray Actuation signal will reset, then Containment Spray Actuation WILL reinitiate after the Reset Spray Actuation pushbuttons are released.

Answer c **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 026000A405 A4.05 **RO Value:** 3.5 **SRO Value:** 3.5 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Containment Spray System 026

KA Statement: Ability to manually operate and/or monitor in the control room:
Containment spray reset switches

Explanation of Answers: 55.41.b(7) Containment Spray actuation relays have retentive memory, which allows relays to be manually reset with an actuation signal still present. So even though containment pressure remains above the Containment Spray actuation setpoint of 15 psig, the actuation signal can be reset. The Containment Spray pumps and Spray valves CS2, CS14, CS16, and CS17 do NOT reposition to their normal positions. The CS14 is normally open with power removed. The remaining CS valves are normally shut and open upon the CS signal, and cannot be closed until the spray actuation signal is reset.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
RPS Logics Safeguards Actuation Signals	221057			23

L.O. Number
CSPRAYE008

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 45

Given the following conditions:

- Salem Unit 1 was operating at 100% power when a LOCA occurred.
- A manual reactor trip and manual SI were initiated.
- When the Main Generator output breakers opened, a loss of off-site power occurred.
- 1A vital bus locked out on bus differential.

Which of the following identifies which Hydrogen Recombiners can be started when directed by procedure if required?

a. 11 ONLY.

b. 12 ONLY.

c. 11 AND 12.

d. Neither Hydrogen Recombiner is available.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 028000K201 K2.01 **RO Value:** 2.5* **SRO Value:** 2.8* **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Hydrogen Recombiner and Purge Control System 028

KA Statement: Knowledge of bus power supplies to the following:
Hydrogen recombiners

Explanation of Answers: 55.41.b(9). Hyd recomb are powered from 1A and 1B 460 volt vital buses, which are powered from their respective 4KV vital buses. With 1A bus locked out on diff, 1A 460 will not have power. Only 12 is available.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
1A Aux Building 460V Bus One line	601231			17
1B Aux Building 460V Bus One line	601232			18

L.O. Number
CONTMTE004

Objectives

Material Required for Examination

Question Source: Previous 2 NRC Exams **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 46

Given the following condition:

Unit 2 Spent Fuel Pool makeup is required when responding to OHA C-35, SFP LVL LO, which has been determined to be caused by normal evaporation over the course of several months.

IAW S2.OP-SO.SF-0001, Fill and Transfer of the Spent Fuel Pool, which of the following is the **LEAST** preferred source of makeup water to the Spent Fuel Pool, assuming all sources of makeup are available?

a. CVCS HUT.

b. Demineralized Water.

c. Primary Water Storage Tank.

d. Refueling Water Storage Tank.

Answer	d	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	033000K105	K1.05	**RO Value:**	2.7*	**SRO Value:**	2.8*	**Section:**	SYS	**RO Group:**	2	**SRO Group:**	2	55.43

System/Evolution Title Spent Fuel Pool Cooling System | 033

KA Statement: Knowledge of the physical connections and/or cause-effect relationships between Spent Fuel Pool Cooling System and the following:
RWST

Explanation of Answers: 55.41(13) The sources of water are listed, in order of preference, in Prerequisite 2.3 of procedure. RWST is last.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Fill and Transfer of the Spent Fuel Pool	S2.OP-SO.SF-0001			20

L.O. Number | Objectives

SFP000E010

Material Required for Examination

Question Source: Facility Exam Bank | **Question Modification Method:** Significantly Modified | **Used During Training Program**

Question Source Comments 44165 changed from preferred to least preferred to match K/A

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 47

Which of the following describes an interlock on the **Unit 2** Spent Fuel Handling Crane?

a. A high load weight cutoff prevents lifting and movement of loads >2200 lbs. over the spent fuel pool.

b. A high radiation signal on 2R32A, Fuel Handling Crane, will lockout ALL crane motion to prevent excessive radiation dose rate on bridge.

c. A bridge over travel limit will stop the bridge from contacting the hard stops even if the direction button is held down, to prevent damage to the bridge.

d. A high radiation signal on 2R5, FHB-SFP, OR 2R9, FHB-New Fuel Storage Area prevents upward crane movement to prevent excessive radiation does rate on bridge.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 034000A302 A3.02 **RO Value:** 2.5* **SRO Value:** 3.1 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Fuel Handling Equipment System 034

KA Statement: Ability to monitor automatic operations of the Fuel Handling Equipment System including:
Load limits

Explanation of Answers: 55.41.b(10,12) The overload cutout is provided to prevent lifting loads >2200 lbs. If they can't be lifted, then they can't be moved over the spent fuel in the pool IAW TR 3.9.7, and is tested IAW TR 4.9.7. B is incorrect because downward crane movement is still available to lower the fuel bundle. C is incorrect because on Unit 2, the proximity switches only transfer the crane to slow, but movement will continue if the direction PB is continued to be depressed. On unit one, it stops movement. The 2R5 and 2R9 are not interlocked with the crane, but are interlocked with FHB ventilation. Technical Requirements Manual 3.9.7 surveillance TR 4.9.7 applicable with fuel in the SFP.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Spent Fuel Pool Manipulations	S2.OP-IO.ZZ-0010			33
Technical Requirements Manual	SGS-TRM	3.9.7	TRM 3/4	2

L.O. Number
IOP010E004

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments 111924

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 48

Unit 2 is performing a plant startup to full power.

Predict the Steam Generator Narrow Range Levels for the following TURBINE power levels.

Assume a normal power ascension.

10% Turbine Power = ____% NR level.

60% Turbine Power = ____% NR level.

a. 38.5%. 46%.

b. 38.5%. 44%.

c. 33%. 46%.

d. 33%. 44%.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 035000A101 A1.01 **RO Value:** 3.6 **SRO Value:** 3.8 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Steam Generator System 035

KA Statement: Ability to predict and/or monitor changes in parameters associated with operating the Steam Generator System controls including: S/G wide and narrow range level during startup, shutdown, and normal operations

Explanation of Answers: 55.41(4) SG NR level is programmed so that as Turbine power rises from 0-20% (Turbine steamline inlet pressure) it rises from 33-44%. On Unit 1 only, 20-100 it rises from 44-48. A is incorrect but plausible if candidate thinks programmed level is based Turbine power but uses Unit 1 ramp from 20-100 (calculated at 60% power. B is correct. C is incorrect but plausible if it is thought that that the ramp is from 20-100% and uses Unit 1 ramp. D is incorrect but plausible if it is thought that that the ramp is from 20-100%.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Advanced Digital Feedwater Control System	NOS05ADFWCS-08		14,17,22	8

L.O. Number

CN&FDWE008

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Significantly Modified **Used During Training Program**

Question Source Comments 125789 Modified "reactor power levels" to "turbine power levels". Changed correct answer from d to b.

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 49

Given the following conditions:

- Unit 1 is operating at 1x10-8 Amps.
- A small (0.1%) steam leak occurs in containment on a single SG.

Which of the following describes how this will affect Rx power with NO operator action?

Rx power will rise....

a. until it stabilizes after reaching the POAH.

b. until the power excursion is terminated by a Rx trip on OT/DT.

c. initially, then stabilize at a power slightly above 1X10-8 Amps.

d. until the power excursion is terminated by a Rx trip on high power (low range).

Answer	a	Exam Level	R	Cognitive Level	Application	Facility:	Salem 1 & 2	ExamDate:	12/19/2016

KA:	039000K508	K5.08	RO Value:	3.6	SRO Value:	3.6	Section:	SYS	RO Group:	1	SRO Group:	1	55.43

System/Evolution Title Main and Reheat Steam System 039

KA Statement: Knowledge of the operational implications of the following concepts as they apply to the Main and Reheat Steam System:
Effect of steam removal on reactivity

Explanation of Answers: 55.41.b(1,5) At 1x10-8 Amps, the reactor is exactly critical. The steam leak will cause Rx power to rise. Before reaching the Point of Adding heat, reactor power will not have a temperature coefficient feedback. The power rise will be terminated AFTER reaching the POAH, as the negative reactivity added will offset the positive reactivity from the steam leak. The 2 reactor trips are plausible if it is thought that power will continue to rise uncontrollably, as the OT/DT trip is always active (not re-instated above 10% power as other trips are), and the high power (lo range) trip would be active since it is only blocked after reaching 10% power.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision

L.O. Number
REACOEE001

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 50

Given the following conditions:

- Unit 2 is operating at 46% power steady state.
- The Main Generator trips due to a ground fault.

Which of the following describes how the Main Generator trip affects the plant with NO operator action?

a. A reactor trip will NOT occur. Main steam dumps will open and control rods will insert. Tavg will stabilize ~547 °F.

b. A reactor trip will NOT occur. Main steam dumps will open, and control rods will insert. Tavg will stabilize ~552 °F.

c. A reactor trip will occur due to the Main Turbine tripping from the Generator Protection trip. The crew will perform TRIP-1 stabilization and diagnostics actions, then transition to TRIP-3.

d. A reactor trip will occur due to the Main Turbine tripping from the Generator Protection trip. The crew will perform Immedate Actions of TRIP-1 then transition to TRIP-2.

Answer	b	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	045000K301	K3.01	**RO Value:**	2.8	**SRO Value:**	3.1	**Section:**	SYS	**RO Group:**	2	**SRO Group:**	2	55.43

System/Evolution Title Main Turbine Generator System | 045

KA Statement: Knowledge of the effect that a loss or malfunction of the Main Turbine Generator System will have on the following:
Remainder of the plant

Explanation of Answers: 55.41.b(4,5,6,7) The generator trip causes a MT trip, but at less than P-9 (49% Rx power), a Reactor trip will NOT occur. With no operator action, rods will insert in auto to lower Tave. The steam dumps will open in Average Temperature Control Load Reject Control Mode. With the reference signal being derived from Main Turbine Steamline inlet pressure, and the Main Turbine is tripped, the reference signal is set to 552°F. (There is a 5°F deadband in which the output to the steam dumps is zero.) Tave rises during the load rejection, and the dumps will modulate to maintain Tave 5 deg above no load Tref of ~547. Control rods insert in auto until

Rod Control T-error is <1.0°F from program, which in this case will be 547-548 since its program signal is based on Turbine Steamline inlet pressure, which is zero.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Main Steam Dumps Lesson Plan	NOS05STDUMP-12		35-36	12

L.O. Number
STDUMPE008

Objectives

Material Required for Examination

Question Source:	Facility Exam Bank	**Question Modification Method:**	Concept Used	**Used During Training Program**

Question Source Comments 57628 (100% power question changed initial power to 46% and added MT trip as the load reject vs SGFP trip)

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: RO 51

Given the following conditions:

- Unit 2 is at 35% power during a plant shutdown performed IAW S2.OP-IO.ZZ-0004, Power Operation.
- 21 SGFP is manually tripped as directed by S2.OP-SO.CN-0002, Steam Generator Feed Pump Operation.
- 24BF19 fails open, and 24 SG NR level rises and remains >67%.

Which of the following describes the effect of this failure, and how will the crew respond?

a. NO AFW pumps start immediately when 22 SGFP trips. If total AFW flow is <22E4 lbm/hr when checked in TRIP-2, the crew will transition to FRHS-1. FRHS-1.

b. NO AFW pumps start immediately when 22 SGFP trips. If total AFW flow is <22E4 lbm/hr when checked in TRIP-2, the crew will start AFW pumps to establish at least 22E4 lbm/hr AFW flow. Flow

c. ONLY 21 and 22 AFW Pumps automatically start when 22 SGFP trips. If total AFW flow is <22E4 lbm/hr when checked in TRIP-2, the crew will transition to FRHS-1. FRHS-1.

d. ONLY 21 and 22 AFW Pumps automatically start when 22 SGFP trips. If total AFW flow is <22E4 lbm/hr when checked in TRIP-2, the crew will start AFW pumps to establish at least 22E4 lbm/hr AFW flow. Flow

Answer	Exam Level	Cognitive Level	Facility	ExamDate
b	R	Application	Salem 1 & 2	12/19/2016

KA		RO Value	SRO Value	Section	RO Group	SRO Group	
059000A201	A2.01	3.4*	3.6*	SYS	1	1	55.43

System/Evolution Title: Main Feedwater System | 059

KA Statement: Ability to (a) predict the impacts of the following on the Main Feedwater System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:

Feedwater actuation of AFW system

Explanation of Answers: 55.41.b(4) The high SG NR level in one SG initiates Feedwater Isolation. This trips both SGFPs. Normally, a trip of both SGFPs would start the MDAFW pumps ONLY (21 and 22), but this does NOT occur when the SGFP signal is caused by a FW Isolation. With power initially at 35%, SG NR levels will remain above the AFW pump auto start level of 9%, so NO AFW pumps will start IMMEDIATELY, they will ALL start when SGNR levels are <9%. If levels did not shrink low enough, then when asked in TRIP-2 step 3 for AFW flow >22E4 lbm/hr, the NO path has operators start 21-23 AFW pumps as necessary to establish >22E4th. The

FRHS transition is plausible because CFSTs would have become active after the transition out of TRIP-1.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Fluency List	NOS05FLUNCY-09		14	9
Reactor Trip Response	2-EOP-TRIP-2		Sh 1	30

L.O. Number: CN&FDWE006

Objectives

Material Required for Examination:

Question Source: New | **Question Modification Method:** | **Used During Training Program**

Question Source Comments:

Comment:

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 52

Which of the following identifies how over-cooling of the RCS is prevented on an uncomplicated manual Rx trip from 100% power?

a. P-10 actuates.

b. P-12 actuates.

c. Feedwater Isolation.

d. Feedwater Interlock.

Answer	d	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	059000K105	K1.05	**RO Value:**	3.1*	**SRO Value:**	3.2	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	☐

System/Evolution Title Main Feedwater System 059

KA Statement: Knowledge of the physical connections and/or cause-effect relationships between Main Feedwater System and the following: RCS

Explanation of Answers: 55.41.b(7)Feedwater interlock actuates when 3/4 RCS Tavgs <554°F and at least one Reactor Trip and associated bypass breaker open. This shuts the BF19's and BF40 Feed Reg Valves. Feedwater Isolation occurs when 2/3 SG NR levels on 1/4 SG's reaches 67% OR on a SI signal, and shuts BF19s, 40s, 13s, and trips the SGFPs. On an uncomplicated Rx trip this will not occur. P-10 is 3/4 PRNIs <10% power, and blocks the low power Rx trips. P-12 is 3/4 RCS Tavgs <543°F, and will shut the Steam Dump valves. On an uncomplicated trip, steam dumps will modulate to control Tavg at 547°, and temp will not reach 543

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Reactor Trip Response	2-EOP-TRIP-2		9	30

L.O. Number
CN&FDWE006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments 159006

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 53

Which of the following describes the design purpose of 21-23AF141, AUX FDWTR PMP INLET AUTO VENT RLSE valves?

a. To facilitate filling lines upon aligning alternate water sources to suction header.

b. To facilitate draining lines upon return from alternate water source alignment to the suction header.

c. To prevent AFW line voiding and water hammer upon AFW pump start when aligned to normal water source.

d. To prevent AFW line temperature gradient if backleakage to AFW from Main Feed occurs with AFW pumps secured.

Answer	a	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	061000K505	K5.05	**RO Value:**	2.7	**SRO Value:**	3.2	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43

System/Evolution Title Auxiliary / Emergency Feedwater System 061

KA Statement: Knowledge of the operational implications of the following concepts as they apply to the Auxiliary / Emergency Feedwater System: Feed line voiding and water hammer

Explanation of Answers: 55.41.(b)7,8. Alternate water sources can be aligned to the AFW pump suction header through the normally jacked closed AF52 Alternate Suction Valves from Demin water, Fresh Water / Fire Protection, and Service Water. During performance of S2.OP-SO.AF-0001, Auxiliary Feedwater System Operation, Section 5.10, Alternate Sources Alignment, operators are instructed to ensure open the Auto Vent Isolation Valves 21-23AF140. This is to allow the alternate suction line to fill and vent through the auto vents, which close when the line is vented.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Auxiliary Feedwater System Operation	S2.OP-SO.AF-0001		65	40
Auxiliary Feedwater System Lesson Plan	NOS05AFW000-15		55	15

L.O. Number
AFW000E004
AFW000E013

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 54

Given the following conditions:

- Unit 1 is operating at 100% power, steady state.
- The Pressure Override circuit for 11 AFW pump fails, effectively removing the Pressure Override circuit from 11 AFW pump control.

Which of the following describes the effect this failure will have on the AFW system?

a. 11AF21 and 12AF21 will remain shut until manually opened if required.

b. 13AF21 and 14AF21 will remain shut until manually opened if required.

c. 11AF21 and 12AF21 will open to the position corresponding to current AF21 demand.

d. 13AF21 and 14AF21 will open to the position corresponding to current AF21 demand.

Answer d **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 061000K601 K6.01 **RO Value:** 2.5 **SRO Value:** 2.8* **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Auxiliary / Emergency Feedwater System 061

KA Statement: Knowledge of the of the effect of a loss or malfunction on the following will have on the Auxiliary / Emergency Feedwater System: Controllers and Positioners

Explanation of Answers: 55.41.b(4,7) The Pressure Override circuit provides runout protection for the AFW pumps by maintaining the AFW pump discharge valves associated with each pump shut until AFW pump reaches a certain pressure, then allows them to open. So during normal standby operation, this circuit keeps the AF21s shut even though normal demand for the valves is set at ~98%. With this circuit removed from affecting the associated AFW21 valves, they will open to current demand. The 11 AFW pump feeds 13 and 14AF21s only, while the 12 AFW pump feeds 11 and 12 AF21s.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
AFW System Lesson Plan	NOS05AFW000-15		27-29	15

L.O. Number

AFW000E008

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 55

Given the following conditions:

- Unit 2 operators are responding to a valid safety injection signal.
- Safety Injection has been RESET.
- ALL SECs have been RESET.
- ALL EDGs are running unloaded.
- 4KV Vital Busses 2A and 2C are aligned to 23 SPT.
- 4KV Vital Bus 2B is aligned to 24 SPT.

Subsequently, 24 SPT experiences a failure causing its secondary voltage to drop and stabilize at 3600 volts.

Which of the following describes the effect of this failure on 2B Vital Bus?

2B Vital Bus will...

a. fast transfer to 23 SPT.

b. remain loaded onto 24 SPT.

c. be energized by its EDG in ACCIDENT loading.

d. be energized by its EDG in BLACKOUT loading.

Answer d **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 062000K302 K3.02 **RO Value:** 4.1 **SRO Value:** 4.4 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title A.C. Electrical Distribution 062

KA Statement: Knowledge of the effect that a loss or malfunction of the A.C. Electrical Distribution will have on the following: ED/G

Explanation of Answers: 55.41.b(7,8) - A is incorrect because 3600V is not <70%, so the vital bus transfer relay will not energize. B is incorrect because the sustained degraded voltage relay will cause a UV signal to be generated for that bus.
C is incorrect because after SI is RESET, the SEC will not actuate in MODE III
D is correct because 3600V is below the setpoint for the degraded voltage (95.%) relays. When these relays actuate then 2B SEC will strip 2B 4KV bus from offsite power and load the bus in MODE II* (Single Bus UV)

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
4160 Electrical System	NOS054KVRLY-00		23-28	0

L.O. Number
4KVAC0E006

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program**

Question Source Comments 45175

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 56

While operating at 100% power, the 1A 28 VDC bus becomes deenergized.

Which of the following identifies an effect this failure will have?

a. 13 AFW pump will start.

b. OHA G-6, 11 SGFP TRBL will annunciate.

c. Control Room Emergency Lighting will light.

d. All indication on 1RP4 status panel will be lost.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 063000K302 K3.02 **RO Value:** 3.5 **SRO Value:** 3.7 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title D.C. Electrical Distribution 063

KA Statement: Knowledge of the effect that a loss or malfunction of the D.C. Electrical Distribution will have on the following:
Components using dc control power

Explanation of Answers: 55.41.b(7) A is incorrect but plausible since loss of 125VDC control power to MS132 steam supply valve will start cause it to fail open and start 13 AFW pump. B is incorrect since 115 VAC power is supplied to alarm panel functions. C is incorrect because 125VDC supplies control room emergency lighting. The purpose of the 28VDC Control Power System is to provide DC electrical power under normal, transient, and accident conditions to:
a.Switchgear
b.Main Control Room Control Console Bezel controls
c.Annunciators
d.Vital Instrument Buses
e.Non safety–related equipment
f.Status panel RP4 in the main control room

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
DC Electrical System Lesson Plan	NOS05DEELEC-09		10-11	9
#1 Unit 28 VDC One Line Drawing	211357		Sh 1	15

L.O. Number
DCELECE008

Objectives

Material Required for Examination	

Question Source:	New	Question Modification Method:		Used During Training Program	☐

Question Source Comments	

Comment

RO SkyScraper	SRO Skyscraper	RO System/Evolution List	SRO System/Evolution List	Outline Changes

Question Topic RO 57

Given the following conditions:

- 2A EDG is running and develops a fuel oil leak.
- 2A EDG Fuel Oil Day Tank level lowers below the start setpoint for both the REGULAR (33") and BACKUP (27") EDG Fuel Oil Transfer Pumps.
- The leak is isolated, and 2A EDG Fuel Oil Day Tank level is recovering.

With 2A EDG Fuel Oil Day Tank level now at 35", which of the following identifies which EDG Fuel Oil Transfer Pumps will be running, if any?

a. NEITHER.

b. BACKUP ONLY.

c. REGULAR ONLY.

d. BOTH REGULAR AND BACKUP.

Answer	d	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	064000K103	K1.03	**RO Value:**	3.6	**SRO Value:**	4.0	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	☐	

System/Evolution Title Emergency Diesel Generators 064

KA Statement: Knowledge of the physical connections and/or cause-effect relationships between Emergency Diesel Generators and the following: Diesel fuel oil supply system

Explanation of Answers: 55.41(7) Regular starts at 33" and Backup at 27", as stated in stem. Once started, they BOTH run until the FULL level is reached (44"). The Backup does NOT turn off when the normal level is reached (33"). It is not require to know the exact setpoint at which the pumps turn off, only that it is substantially higher that when the Regular pump turns on, and that the backup pump will turn off at the same level as the Regular pump.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
NOS05EDG000-12	EMERGENCY DIESEL GENERAT		44	12

L.O. Number
EDG000E007

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program** ☐

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 58

Given the following conditions:

- Unit 1 is operating at 100% power.
- 1A EDG starting air receiver 11A is C/T.
- 11B EDG starting air compressor is C/T.
- A Loss of Off-Site Power occurs.
- 1A EDG trips shortly after starting.
- 11A starting air compressor will NOT run.

By design, what is the minimum number of subsequent 1A EDG start attempts available with the air start system in this configuration?

a. 2

b. 3

c. 4

d. 5

Answer	a	**Exam Level**	R	**Cognitive Level**	Memory	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	064000K607	K6.07	**RO Value:**	2.7	**SRO Value:**	2.9	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	

System/Evolution Title Emergency Diesel Generators | 064

KA Statement: Knowledge of the of the effect of a loss or malfunction on the following will have on the Emergency Diesel Generators:
Air receivers

Explanation of Answers: 55.41.b(8) Salem FSAR section 9.5.6, "Each receiver is sized to hold sufficient air for three cold diesel starts". Each EDG has 2 air receivers, but with one C/T, there will be 2 starts left after 1 attempt at starting.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
EDG Lesson Plan	NOS05EDG000-12		48-49	12
Diesel Engine Auxiliaries	205241		Sh 1	42

L.O. Number

EDG000E014

Objectives

Material Required for Examination

Question Source:	Facility Exam Bank	**Question Modification Method:**	Editorially Modified	**Used During Training Program**

Question Source Comments 80450

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 59

Given the following conditions:

- Unit 2 is performing a normal Liquid Release from 21 CVCS Monitor Tank via 22 SW header to Unit 1 CW IAW S2.OP-SO.WL-0001, Release of Radioactive Liquid Waste from 21 CVCS monitor Tank.
- The Radwaste Overboard Discharge Flow Recorder is OPERABLE.
- The 2R18 and the 2R41D rad monitors are OPERABLE.
- After commencing the release, the operator is compiling the Initial Data IAW appropriate attachment.
- When recording the 2R18 reading, it reads 10E6 cps.
- The 2R18 red alarm light is lit on the 104 panel.
- The 2WL51 indicates OPEN in the control room.

Which of the following describes what these indications mean, and how the operating crew should proceed?

a. The 2WL51 should have automatically closed on the valid 2R18 high radiation alarm. The NEO should shut 2WL51 locally.

b. The 2WL51 should have automatically closed on the valid 2R18 high radiation alarm. The NCO in the control room should shut 2WL51.

c. 2R18 has failed high. Verify the flow recorder and 2R41D are OPERABLE, block the R-18 input to the 2WL51 on 2RP1, and continue the liquid release.

d. The 2R18 reading is well below the expected Hi Rad Alarm setpoint. Verify the flow recorder and 2R41D are OPERABLE, block the R-18 input to the 2WL51 on 2RP1, and continue the liquid release.

Answer b **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 068000K610 K6.10 **RO Value:** 2.5 **SRO Value:** 2.9 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Liquid Radwaste System 068

KA Statement: Knowledge of the of the effect of a loss or malfunction on the following will have on the Liquid Radwaste System: Radiation monitors

Explanation of Answers: 55.41.b(11,13)6.82E5 cps is the ALARM setpoint which will automatically shut the 2WL51 (S2.IC-CC.RM-0028) The current reading is above the setpoint at which the 2WL51 should have automatically shut, and the NCO should shut the valve remotely. Additionally, S2.OP-SO.WL-0001 for releasing the tank, Step 5.5.9, says if the 2R18 alarms, then the NEO is to inform the NCO to shut the 2WL51. There is no provision for closing the valve locally.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
RELEASE OF RADIOACTIVE LIQUID WASTE	S2.OP-SO.WL-0001		13	27

L.O. Number WASLIQE005

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program**

Question Source Comments 83676

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 60

When performing a Source Check of a Process radiation monitor, which of the following describes how to initiate the Source Check, and why?

Activate the Source Check......

a. for 30 seconds or less to prevent solenoids from overheating.

b. just until indication of rising level is detected. This will prolong source life.

c. for 30 seconds or less to prevent unwanted automatic action on high radiation level.

d. just until indication of rising level is detected. This will ensure radiation level outside monitor remains less than detectable.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 073000G132 2.1.32 **RO Value:** 3.8 **SRO Value:** 4.0 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Process Radiation Monitoring System 073

KA Statement:
Ability to explain and apply all system limits and precautions.

Explanation of Answers: 55.41.b(11) Both the system operating and surveillance procedures contain precautions limiting check source operation to 30 seconds or less. B is incorrect but plausible because it's a true statement , but it is not the reason. C is incorrect but plausible since there are some monitors whose alarm action functions are not automatically blocked during a source check. D is incorrect because source check operation is not intended to raise rad levels outside component.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Radiation Monitors - Check Sources	S1.OP-ST.RM-0001		3	31
Radiation Monitoring System Operation	S1.OP-SO.RM-0001		4	38

L.O. Number
RMS000E006

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 61

Given the following conditions:

- Unit 2 is operating at 100% power during a summer heat wave.
- 21, 24, and 26 SW pumps are in service.
- SW header pressure is 106 psig.
- The 2SW308 #2 Bay Overpressure Protection valve slowly fails open.

Which of the following describes how the plant will respond with NO operator action?

OHA's B-13 21 SW HDR PRESS LO and B-14 22 SW HDR PRESS LO will annunciate at 1) _______ psig, and the standby SW pump will auto start at 2) _____ psig.

a. 1) 99.5
2) 95.5.

b. 1) 99.5
2) 99.5.

c. 1) 105.0
2) 95.5.

d. 1) 105.0
2) 99.5.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 076000K402 K4.02 **RO Value:** 2.9 **SRO Value:** 3.2 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Service Water System 076

KA Statement: Knowledge of Service Water System design feature(s) and or interlock(s) which provide for the following:
Automatic start features associated with SWS pump controls

Explanation of Answers: 55.41.b(7) Low pressure 99.5, auto start 95.5

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Overhead Annunciator Window B	S2.OP-AR.ZZ-0002		28-29	36

L.O. Number
SWBAYSE009

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 157024

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: RO 62

Unit 2 ECAC is running loaded for testing IAW S2.OP-PT.CA-0001, Emergency Control Air Compressor Functional Test.

Which of the following would have an effect on Control Air Header pressure?

a. 2A 4KV to 460V breaker 2A4D trips.

b. 2C 4KV to 460V breaker 2C4D trips.

c. 2E 4KV to 460V breaker 2E6D trips.

d. 2H 4KV to 460V breaker 2H5D trips.

Answer: b **Exam Level**: R **Cognitive Level**: Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 078000K202 K2.02 **RO Value:** 3.3* **SRO Value:** 3.5* **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title: Instrument Air System 078

KA Statement: Knowledge of bus power supplies to the following:
Emergency air compressor

Explanation of Answers: 55.41.b(8) Unit 2 ECAC is powered from 2C 460 volt bus. Station Air Compressors are powered from Group buses.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of 2C 460/230V Vital Bus	S2.OP-AB.460-0003		7	7

L.O. Number: CONAIRE005

Objectives

Material Required for Examination:

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments:

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 63

Which of the following describes Station Blackout (SBO) Compressor operation and its connection to the Control Air System during a station blackout?

The SBO Compressor ___(1)____ automatically start on a loss of all station power. After starting, the SBO ___(2)___ the 1A and 2A Control Air Headers.

a. 1) will 2) must be manually aligned to

b. 1) will 2) automatically supplies

c. 1) will NOT 2) must be manually aligned to

d. 1) will NOT 2) automatically supplies

Answer c **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 079000K401 K4.01 **RO Value:** 2.9 **SRO Value:** 3.2 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Station Air System 079

KA Statement: Knowledge of Station Air System design feature(s) and or interlock(s) which provide for the following:
Cross-connect with IAS

Explanation of Answers: 55.41.b(8) The SBO Compressor is required to be isolated from, and independent of plant safety related equipment except when required during operation during a blackout or other situations when there is a total loss of Control Air. The SBO is manually started and manually aligned to the CA headers 1A and 2A.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
SBO Diesel Air Compressor	SC.OP-SO.CA-0001			14

L.O. Number
STAAIRE013

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 64

Given the following conditions:

- 2B EDG is in service paralleled with the 2B 4KV vital bus for a scheduled surveillance run.
- A fire occurs in 2B EDG room, and the fire detector in the room senses the fire.

Which of the following identifies how the Fire Protection system responds, and how the EDG operator should respond?

a. The EDG CO2 fire suppression system must be manually activated by the operator as they exit the EDG vestibule area.

b. The EDG Halon fire suppression system must be manually activated by the operator as they exit the EDG vestibule area.

c. The EDG CO2 fire suppression system will automatically discharge after a 13 second delay, the operator must immediately exit the 2B EDG area.

d. The EDG Halon fire suppression system will automatically discharge after a 13 second delay, the operator must immediately exit the 2B EDG area.

Answer a **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 086000K504 K5.04 **RO Value:** 2.9 **SRO Value:** 3.5* **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43

System/Evolution Title Fire Protection System 086

KA Statement: Knowledge of the operational implications of the following concepts as they apply to the Fire Protection System:
Hazards to personnel as a result of fire type and methods of protection

Explanation of Answers: 55.41.b(7) The EDGs are protected by a CO2 fire suppression system. This previously AUTOMATIC system was permanently changed to a MANUAL activation system under Salem DCP 80115237. The Fire alarm will alert the operator to the fire, at which time they exit the EDG room/control room to activate the fire suppression system from the vestibule area. There is a 13 second after activation before the system dumps. The Halon distracters are plausible because Salem has other vital areas which have Halon fire suppression systems. This meets the K/A as the operator must know if the dump automatically occurs (immediately required to exit area) or if there is other action which must be taken, as well as knowing the methods of protection (CO2 vs Halon)

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
EDG Lesson Plan	NOS05EDG000-12			12

L.O. Number
FIRPROE006

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper	SRO Skyscraper	RO System/Evolution List	SRO System/Evolution List	Outline Changes

Question Topic RO 65

Given the following conditions:

- Unit 2 was operating at 100% power when a steam leak upstream of 22MS167 occurred.
- The Rx was tripped and a MSLI performed successfully.
- Operators have transitioned out of EOP-TRIP-1.
- The PO is attempting to open 21-24SS94s, SG B/D Sample Valves, but they will not open.
- SGBD sample isolation bypass has been RESET.

Of the following, which identifies the reason the valves won't open?

a. 22 SG NR level is <9%. 9%.

b. CA330s have not been reopened.

c. Both SGFPs have trip signals locked in.

d. Either Train A or Train B Phase A isolation failed to reset when its reset PB was depressed.

Answer	d	**Exam Level**	R	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	103000A404	A4.04	**RO Value:**	3.5*	**SRO Value:**	3.5*	**Section:**	SYS	**RO Group:**	1	**SRO Group:**	1	55.43	☐

System/Evolution Title Containment System | 103

KA Statement: Ability to manually operate and/or monitor in the control room:
Phase A and phase B resets

Explanation of Answers: 55.41.b(7) For a trip and SI due to a single faulted SG (unisolable) the flow path will go from TRIP-1 to LOSC-1. The SI will NOT have been reset in TRIP-1, nor will it be reset in LOSC-1. The SGBD sample isolation reset will be performed in LOSC-1 (step 6.1) in order to open the SS94's. The step prior to that is RESET PHASE A. This is due to the fact that the blowdown isolation bypass only bypasses the lo-lo level input into the AFW auto start circuit, which closes the SS94's. If the Phase A hasn't been reset, the 94s can not be reopened. SS94s supplied air from outside cont

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
RPS AFW Startup Logic Diagram	221064			8
SS94 Loop Diagram	621216-1			1

L.O. Number

RXPROTE019

Objectives

Material Required for Examination

Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐

Question Source Comments 62124

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 66

During shift turnover, the on-coming RO notices an OHA with reflash capability has 1 (one) piece of red translucent tape diagonally across its window box.

Which of the following describes what the status of this OHA is?

a. The alarm is inoperable, and will not annunciate under any circumstances.

b. The alarm is identified for heightened awareness, and has all functionality present.

c. The alarm has at least one, but not all, inputs disabled, and may not be a reliable source of information.

d. The alarms reflash capability has been defeated such that if already in alarm, a second alarm will not be annunciated.

Answer c **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G101 2.1.1 **RO Value:** 3.8 **SRO Value:** 4.2 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement: Knowledge of conduct of operations requirements.

Explanation of Answers: 55.41.b(10) The single piece of tape is placed on the OHA to alert the operator that the alarm is not a reliable source of information. 2 pieces of red tape in a "X" signifies that the entire OHA window is INOPERABLE. Additionally, if one or more inputs to a multiple input annunciator are inoperable, then red tape should be placed diagonally across the annunciator window. A is incorrect because the window can still alarm from an operable input, since it is not ("X'd). B is incorrect because the tape signifies that there is something wrong with it, so it does NOT have full functionality. C is correct as per above. D is incorrect because the reflash capability is not defeated, and as long as a second valid input comes in with one already in, the reflash capability of the alarm will cause it to annunciate.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Operator Burdens Program	OP-AA-102-103-1001		8	2

L.O. Number

CONDOPE005

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 67

Given the following conditions:

- Unit 2 is operating at 60% power.
- Power was reduced when 22 SGFP tripped 2 days ago, and has remained at this level for 2 days.
- Operators are preparing to start a Main Turbine power ascension using dilution and automatic rod control to maintain Tavg and AFD on program.
- Tavg-Tref deviation is 0°F.

IAW OP-AA-300, Reactivity Management, how should the power ascension be started?

Assume this is a normal power ascension with all equipment available.

a. The crew concurrently initiates the Main Turbine load ascension and a RCS dilution.

b. The crew initiates a RCS dilution. As soon as the dilution is in progress the Main Turbine load ascension is initiated.

c. The crew initiates a RCS dilution and waits until a RCS temperature rise is detected. Then the Main Turbine load ascension is initiated.

d. The crew initiates the Main Turbine load ascension and waits until a RCS temperature lowering is detected. Then the RCS dilution is initiated.

Answer c **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G102 2.1.2 **RO Value:** 4.1 **SRO Value:** 4.4 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement: Knowledge of operator responsibilities during all modes of plant operation.

Explanation of Answers: 55.41(5,10) Listed under the responsibilities of the Reactor Operator, page 7, 4.6.5, "Typically, during planned load changes where dilution or boration is required, start with the dilution or boration. The initial effects (RCS temperature change) of the reactivity change should be seen prior to initiating the load change. (PWR)" All the distracters have the correct actions in the wrong order.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Reactivity Management	OP-AA-300		7	7

L.O. Number

CONDOPE005

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 125830

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 68

Which of the following conditions will REQUIRE the suspension of fuel movement in the Unit 2 Rx vessel?

a. Chemistry reports Rx Cavity boron concentration is 2499 ppm.

b. A NEO reports BOTH 100' elevation containment airlock doors are open.

c. The PO depresses Fire Outside Control Room on Unit 2 Control Area Ventilation.

d. Containment Radiation Monitor 2R12A fails causing a Containment Ventilation Isolation signal.

Answer c **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G140 2.1.40 **RO Value:** 2.8 **SRO Value:** 3.9 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement: Knowledge of refueling administrative requirements.

Explanation of Answers: 55.41.b(11) C is correct because operation in the recirculation mode requires suspension of fuel movement. (SO.CAV P&L 3.6.3 When aligned to FIRE OUTSIDE CONTROL AREA (Recirculation Mode),Core Alterations and movement of irradiated fuel is NOT permitted (T/S Bases 3/4.7.6)). D is incorrect because Containment Radiation monitors are not required to be operable for Mode 6 or Fuel Movement or Core Alts per Tech Specs. B is incorrect because the airlock doors are only required to be CAPABLE of being shut, and can be open (S2.OP-ST.CAN-0007, page 8). A is incorrect because the COLR limit for boron concentration is 2139

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs				
Refueling Operations-Containment Closure	S2.OP-ST.CAN-0007		8	28
Control Area Ventilation Operation	S2.OP-SO.CAV-0001		8	41

L.O. Number
REFUELE012

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 110741

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 69

When performing a plant startup, when is MODE 2, STARTUP, entered IAW S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load?

a. When the Rx is declared critical.

b. When Control Bank withdrawal is imminent.

c. When the Reactor Trip Breakers are closed.

d. When Shutdown Bank withdrawal is imminent.

Answer b **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G235 2.2.35 **RO Value:** 3.6 **SRO Value:** 4.5 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement:

Ability to determine Technical Specification Mode of Operation.

Explanation of Answers:

55.41.b(6) Step 4.2.21.3 When the withdrawal of Control Bank "A" is imminent, PERFORM the following:
RECORD time of Mode 2 entry in the Control Room Narrative Log.
UPDATE WCM to Mode 2.
RECORD Date and Time of Control Banks withdrawal in Attachment 3, Technical Specifications Pre-Criticality Surveillance Data, Section A

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Hot Standby to Minimum Load	S2.OP-IO.ZZ-0003		18	43

L.O. Number

IOP003E004

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 155638

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 70

Given the following conditions:

- Unit 2 is operating at 100% power.
- 100% AFD Target is -1.5.
- Between 0100 - 0140 today, 40 AFD penalty minutes were accumulated during a Main Turbine Valve Test which caused a rapid power reduction to be performed when a steam valve could not be re-opened initially.
- At 2300, 22 SGFP trips.
- At 2301, Rx power is 87% and lowering with AFD at -9.1.
- At 2303, Rx power is 66% and the Turbine load reduction is complete, with AFD at -10.1

Assuming power and AFD remain at 66% and -10.1 respectively, which of the following describes the required power reduction the crew must take IAW Salem Tech Specs?

Power must be reduced less than 50% by........

a. 2336.

b. 2338.

c. 2351.

d. 2353.

Answer d **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G239 2.2.39 **RO Value:** 3.9 **SRO Value:** 4.5 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement:

Knowledge of less than or equal to one hour Technical Specification action statements for systems.

Explanation of Answers:

55.41.b(5,6)LCO 3.2.1 action a (for >90% power) and action b (50-90% power) applies. The MT auto runback upon a SGFP trip occurs at 15% per minute. The AFD Target at 100% power (Rev. 18 of Tables) is -1.5, with a band of +6, -9. The target is adjusted for Rx power, so with power at 87%, Target is -1.3, and band of -9 = -10.3, so at 2301 AFD is still within the Target Band. With power at 66%, Target is -0.99, and band -9= -9.99. With indicated AFD at -10.1, AFD is out of band at 2303, and Penalty Minutes accumulate 1 for every minute outside Target Band. With 40 minutes accumulated <24 hours ago, the plant will exceed 60 minutes outside of Target Band at 2303+20=2323. The action for outside of band for 60 minutes of penalty minutes (but still within COLR limits) is to lower power to <50% within 30 minutes. So Rx Thermal power must be <50% with 30 minutes of 2323, which is 2353. The distracters are using the 15 minute action time required for power >90%, or using the 2301 time as the time AFD is out of band (plausible if the -9 of Target Band is incorrectly used as the actual AFD limit) or both.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs		LCO 3.2.1	3/4 2-1	278
Tables	S2.RE-RA.ZZ-0011			248
Salem Core Operating Limits Report	COLR Salem 2		6	6

L.O. Number

FLUNCYE002

Objectives

Material Required for Examination					
Question Source:	New	**Question Modification Method:**		**Used During Training Program**	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 71

Which Emergency Classification results in an automatic extension of annual dose (TEDE) limits for ERO personnel that have an NRC Form-4 on file, and what is the dose limit extended to?

a. SITE AREA EMERGENCY, 5000 mrem.

b. SITE AREA EMERGENCY, 4500 mrem.

c. ALERT, 5000 mrem.

d. ALERT, 4500 mrem.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G304 2.3.4 **RO Value:** 3.2 **SRO Value:** 3.7 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement: Knowledge of radiation exposure limits under normal or emergency conditions.

Explanation of Answers: 55.41.b(12,10) Upon an ALERT declaration, workers with a NRC Form 4 on file have their dose limit raised from 2,000 mrem to 4500 mrem.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
OPERATIONAL SUPPORT CENTER (OSC) RA	NC.EP-EP.ZZ-0304			16

L.O. Number

RADCONE003

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 135687

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 72

A Containment Entry must be made in Mode 1.

When is Radiation Protection Supervisor (RPS) approval required AND what is the limit for the number of personnel that may enter containment?

Assume the Shift Manager has NOT granted permission to exceed the procedural limit

The RPS must give prior approval if the entry is made during ....

a. a power ascension at >5% / hr ONLY, and no more than 10 people may be in containment at one time.

b. any reactor power change at >5% / hr, and no more than 10 people may be in containment at one time.

c. a power ascension at >5% / hr ONLY, and no more than 20 people may be in containment at one time.

d. any reactor power change at >5% / hr, and no more than 20 people may be in containment at one time.

Answer d **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G313 2.3.13 **RO Value:** 3.4 **SRO Value:** 3.8 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement: Knowledge of radiological safety procedures pertaining to licensed operator duties, such as response to radiation monitor alarms, containment entry requirements, fuel handling responsibilities, access to locked high-radiation areas, aligning filters, etc.

Explanation of Answers: 55.41.b(12) SC.SA-ST.ZZ-0001, (Rev. 5) SALEM CONTAINMENT ENTRIES IN MODES 1 THROUGH 4, requires prior RP Supervisor approval to enter containment in Modes 1 or 2 during ANY power change >5% / hr. It also limit entry to 10 people per operable air lock. 2 airlocks = 20 people. See steps 2.4 & 3.2.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
SALEM CONTAINMENT ENTRIES IN MODES	SC.SA-ST.ZZ-0001		3,4	5

L.O. Number

CONTMTE012

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program**

Question Source Comments 159346 MODIFIED ANSWERS FROM "5%" TO "5% / HR."

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 73

Which of the following contains conditions that warrant entry to EOP-LOCA-5 “Loss of Emergency Coolant Recirculation”?

a. RWST LOW LEVEL alarm annunciates with <62% level in Containment Sump.

b. The hot leg recirculation flow path is unavailable due to blockage of flow paths.

c. RWST LOW-LOW LEVEL alarm annunciates requiring all ECCS pumps with suction from the RWST to be stopped.

d. LOCA outside containment is indicated, and both RHR pumps were stopped IAW EOP-LOCA-6, LOCA OUTSIDE CONTAINMENT.

Answer a **Exam Level** R **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G404 2.4.4 **RO Value:** 4.5 **SRO Value:** 4.7 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement:

Ability to recognize abnormal indications for system operating parameters which are entry-level conditions for emergency and abnormal operating procedures.

Explanation of Answers:

55.41.b(10) EOP-LOCA-5 can be entered from 5 places. 1. LOCA-1 step 16 with no RHR pump and associated SJ44 available; 2. LOCA-3 step 2, Containment sump level less than 62%; 3. LOCA-4 step 5, No RHR pumps running; 4. LOCA-6 step 6.2, with RH1,2,26 shut, 21 and 22 RH19 shut, 21 and 22SJ49's shut and RCS pressure not rising; 5. Loop from LOCA-5 step 28 when RWST level is still above LO-LO (1.2') setpoint. Answer is correct because containment sump level should be rising as RWST level is lowering. C is incorrect because it is a condition that would be expected to occur as RWST level dropped during CL recirc as 1 CS pp continued to draw water from RWST. D is incorrect because LOCA 5 only requires RHR pumps to be available, not in service. B is incorrect because there is no provision in LOCA-4 to check if the flowpath is blocked or not. It only has the alignment performed, then return to procedure in effect.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Emergency Recirculation	2-EOP-LOCA-5		1	30
Transfer to Cold Leg Recirculation	2-EOP-LOCA-3		1	30

L.O. Number | **Objectives**

LCA3U1E007

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Editorially Modified	**Used During Training Program**	☐
Question Source Comments	42264 Changed correct answer wording from "RWST inventory is being depleted...." to "RWST LOW LEVEL alarm annunciates...".				

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 74

Unit 2 has entered S2.OP-AB.STM-0001 EXCESSIVE STEAM FLOW due to an extraction steam line rupture. A Reactor trip and MSLI have been performed, and the steam leak has been isolated. Safety Injection was not required and did not initiate.

Which of the following describes how S2.OP-AB.STM-0001 will be used following performance of EOP-TRIP-1, Rx Trip or Safety Injection?

S2.OP-AB.STM-0001.....

a. must be re-entered from point it was left and completed regardless if entry conditions are still met.

b. must be re-entered from the beginning and completed regardless if entry conditions are still met.

c. should be re-evaluated to see if entry conditions are still met. If they are, re-enter AB from the beginning.

d. should be re-evaluated to see if entry conditions are still met. If they are, re-enter the procedure from where it was left at Rx trip.

Answer	c	Exam Level	R	Cognitive Level	Memory	Facility:	Salem 1 & 2	ExamDate:	12/19/2016

KA:	194001G411	2.4.11	RO Value:	4.0	SRO Value:	4.2	Section:	PWG	RO Group:	1	SRO Group:	1	55.43

System/Evolution Title GENERI

KA Statement: Knowledge of abnormal condition procedures.

Explanation of Answers: 55.41.b(10) Step 4.4.1 If a reactor trip occurs while performing an AOP, either intentionally or as directed by the AOP, the AOP should be exited and the EOP entered. If the AOP calls for a trip of the reactor, the action is defined in the same manner as in 4.2.5 (fourth bullet, EOP-TRIP-1) for EOP action steps. There are cases where steps must be completed as directed in the AOP (i.e. TRIP affected RCPs) prior to entering the EOPs. Once this action is completed, then entry into EOPTRIP-1 is directed by procedure steps in the AOP. When the EOPs are exited or additional personnel become available, the entry conditions for the AOP should be re-evaluated to determine if performance of the AOP is still required,(i.e. the reactor trip did not terminate the abnormal condition). If performance is required, the AOP should be re-entered from the beginning.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Use of Procedures	OP-AA-101-111-1003		31	6

L.O. Number

PROCEDE006

Objectives

Material Required for Examination

Question Source:	Facility Exam Bank	Question Modification Method:	Editorially Modified	Used During Training Program	

Question Source Comments 72934 changed from following TRIP-2 to TRIP-1 based on new rev to EOPs

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic RO 75

Given the following conditions:

- Unit 2 is operating at 100% power steady state.
- The unit has achieved "black board" status on the OHA's with NO OHA's currently in alarm.
- An expected OHA is NOT received when opening a panel which provides an OHA.

Which of the following describes the control room response IAW S2.OP-AB.ANN-0001, Loss of Overhead Annunciator System, if it is determined that a complete and total loss of ALL Overhead Annunciators has occurred and cannot be restored within 15 minutes?

a. Initiate alternate alarm check of the OHA system every 15 minutes by opening and closing a RPS or SSPS cabinet door.

b. Initiate a controlled shutdown at <5% per minute to bring the unit off line. Line.

c. Initiate continuous control console walkdown.

d. Trip the Rx and GO TO EOP-TRIP.

Answer c **Exam Level** R **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G432 2.4.32 **RO Value:** 3.6 **SRO Value:** 4.0 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title GENERI

KA Statement: Knowledge of operator response to loss of all annunciators.

Explanation of Answers: 55.41.b(7) With a loss of all annunciators, tripping the Rx or initialing a power reduction are not warranted unless overriding plant status would direct it, not just from the annunciator problem. Initiating continuous walkdowns of control consoles(and P-250, etc.) is directed. With a total loss of OHA's, you don't check functionality by trying to initiate an alarm every 15 minutes, that's performed to verify status if some functionality is present.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Overhead Annunciator System	S2.OP-AB.ANN-0001		14	23

L.O. Number

ABANN1E002

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 155633

Comment

U.S. Nuclear Regulatory Commission
Site-Specific
Written Examination

Applicant Information

Name:	Region: I
Date: 12/19/2016	Facility: Salem 1 & 2
License Level: SRO	Reactor Type: W
Start Time:	Finish Time:

Instructions

Use the answer sheets provided to document your answers. Staple this cover sheet on top of the answer sheets. To pass the examination you must achieve a final grade of at least 80.00 percent overall, with 70.00 percent or better on the SRO-only items if given in conjunction with the RO exam; SRO-only exams given alone require a final grade of 80.00 percent to pass. You have 8 hours to complete the combined examination, and 3 hours if you are only taking the SRO portion.

Applicant Certification

All work done on this examination is my own. I have neither given nor received aid.

____
Applicant's Signature

Results

RO/SRO-Only/Total Examination Values	____ / ____ / ____	Points
Applicant's Score	____ / ____ / ____	Points
Applicant's Grade	____ / ____ / ____	Percent

Senior Reactor Operator Answer Key

1. a
2. d
3. a
4. a
5. a
6. d
7. d
8. a
9. b
10. a
11. b
12. a
13. d
14. c
15. d
16. c
17. b
18. c
19. d
20. b
21. d
22. d
23. b
24. c
25. a

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 76

Given the following conditions:

- Unit 1 is operating at 100% power.
- An automatic Rx trip occurs when a loose wire in Main Power transformer Phase B causes a Main Generator trip.
- The RO performs 1-EOP-TRIP-1 Reactor Trip or Safety Injection immediate actions with no SI required.
- The CRS transitions to 1-EOP-TRIP-2, Reactor Trip Response after verification of immediate actions.
- Only TRIP-1 Immediate Actions were performed prior to transitioning to TRIP-2.

- Upon entry into EOP-TRIP-2, the PO reports:
  - 11 AFW pump has failed to start.
  - 12 AFW pump has started but a malfunctioning Pressure Override circuit is keeping 11AF21 and 12AF21 shut.
  - 13 AFW pump tripped as it was accelerating during start.
  - All SG NR levels are off-scale low.
  - SPDS indicates a Heat Sink Red Path exists.

Which of the following describes the next action the CRS should take?

a. Direct the PO to establish Aux Feedwater flow IAW 1-EOP-TRIP-2, Reactor Trip Response.

b. Direct the PO to establish Main Feedwater flow IAW 1- EOP-TRIP-2, Reactor Trip Response.

c. Immediately transition to 1-EOP-FRHS-1, Loss of Heat Sink, ONLY if 3 of 4 SG WR levels are <32% to establish SG Bleed and Feed.

d. Immediately transition to 1-EOP-FRHS-1, Loss of Heat Sink, based on the SPDS indication of a valid Red Path, and perform a SGFP Prompt Recovery IAW S2.OP-SO.CN-0007, SGFP Prompt Recovery.

Answer	a	**Exam Level**	S	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	000007A202	EA2.02	**RO Value:**	4.3	**SRO Value:**	4.6	**Section:**	EPE	**RO Group:**	1	**SRO Group:**	1	55.43	✔

System/Evolution Title Reactor Trip 007

KA Statement: Ability to determine and interpret the following as they apply to Reactor Trip:
Proper actions to be taken if the automatic safety functions have not taken place

Explanation of Answers: 55.43.b(5) CFST's become active upon exit from EOP-TRIP-1. However, from CFST procedure...."SPDS is not designed to be used as a primary indication, and no actions should be based upon SPDS indications without verification of the primary indications, which are the installed Control Room 1E instruments." The crew ould perform the actions at Step 3 to check total AFW flow >22E4 lbm/hr, and if not, start 11-13 AFW pumps as necessary to establish it, which would include defeating Pressure Override circuit. The Main Feedwater step is AFTER the AFW step. This question is SRO level based on knowing which procedure to use, (do not transition to another procedure) and the actions in that procedure. The FRHS-1 distracters are plausible if incorrect application of CFST usage after transition out of TRIP-1 are applied.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Critical Safety Function Status Trees	EOP-CFST-1			31
Reactor Trip Response	EOP-TRIP-2	Sh 1		30

L.O. Number
TRP001E001
TRP002E005

Objectives

Material Required for Examination					
Question Source:	New	Question Modification Method:		Used During Training Program	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 77

Given the following conditions:

- Salem Unit 2 is in Mode 3, at NOP, NOT.
- All four RCPs are in service.
- 22 RDMG set is in operation for testing, with its motor breaker and generator breaker shut.
- Reactor Trip Breaker "B" is racked in and shut.
- Reactor Trip Breaker "A" and both Reactor Trip Bypass Breakers are open.

Which of the following identifies how Tech Specs will be applied if a RCP were to trip, and the bases for that application?

a. ALL RCPs are required to be operable to provide decay heat removal and to satisfy single failure criteria. A cooldown to Mode 4 IS required.

b. Only TWO reactor coolant loops are required to be operable to provide decay heat removal and to satisfy single failure criteria. A cooldown to Mode 4 is NOT required.

c. ALL RCPs are required to be in operation to provide decay heat removal and ensure mixing, prevent stratification, and produce gradual reactivity changes during boron concentration reductions in the RCS. A cooldown to Mode 4 IS required.

d. Only ONE RCP is required to be in operation to provide decay heat removal and ensure mixing, prevent stratification, and produce gradual reactivity changes during boron concentration reductions in the RCS. A cooldown to Mode 4 is NOT required.

Answer d **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000015G237 2.2.37 **RO Value:** 3.6 **SRO Value:** 4.6 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title Reactor Coolant Pump Malfunctions 015

KA Statement: Ability to determine operability and/or availability of safety related equipment.

Explanation of Answers: 55.43.b(2) The first part of this questions require RO knowledge to determine what constitutes an "energized" rod control system. 3 of the 4 RTB and RTBBs being open is considered de-energized, even with a RDMG set in operation and a single RTB shut. (TS page 3/4 4-2a). With the unit in Mode 3 (rod control deenergized means can't be in Mode 2, and at NOT can't be in Mode 4), LCO 3.4.1.2 requires TWO operable RC loops including RCP, SG, and loop. It also requires ONE loop in operation when rod control is deenergized. With FOUR RCPs (and loops) operable initially, the trip of a single RCP does not require any action based on still meeting the minimum requirement for the LCO. The bases for this LCO is as stated in correct answer. Choice B is partially correct, since the operability portion is correct and the cooldown not required part is also correct. However, the bases does not reflect the reason as being single failure criteria. Single failure criteria refers in this case to the minimum required loops IN OPERATION. (Bases B3/4 page 4-1.)

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs		LCO 3.4.1.2	3/4-2,2a	44/282

L.O. Number

RCS000E009

Objectives

Material Required for Examination	

Question Source:	New	Question Modification Method:		Used During Training Program	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 78

Given the following conditions:

- Salem Unit 2 is operating at 4% power awaiting final approval to enter Mode 1.
- Main steam dumps are in service maintaining Tavg at 548°F.
- 21 SGFP is in service, 22 SGFP is latched and at idle speed.
- All AFW pumps are secured and aligned for normal standby operation.

Subsequently:
- An automatic Rx trip signal is received, but the Rx does not trip.
- All attempts to trip the Rx from the Control Room fail.

Which of the following describes how the CRS should proceed?

Initiate manual rod insertion......

a. then enter 2-EOP-FRSM-1, Response to Nuclear Power Generation, and start both MDAFW pumps.

b. then enter 2-EOP-FRSM-1, Response to Nuclear Power Generation, verify feed flow >44E4 lbm/hr from SGFP's.

c. then continue in 2-EOP-TRIP-1, Reactor Trip or Safety Injection, then TRIP-2, Reactor Trip Response, since FRPs are not in effect in MODE 2.

d. then continue in 2-EOP-TRIP-1, Reactor Trip or Safety Injection, then TRIP-2, Reactor Trip Response, since reactor trip is confirmed with Rx power <5%.

Answer	a	**Exam Level**	S	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA: 000029G244 | 2.2.44 | **RO Value:** 4.2 | **SRO Value:** 4.4 | **Section:** EPE | **RO Group:** 1 | **SRO Group:** 1 | 55.43 ☑

System/Evolution Title Anticipated Transient Without Scram | 029

KA Statement:

Ability to interpret control room indications to verify the status and operation of a system, and understand how operator actions and directives affect plant and system conditions.

Explanation of Answers:

55.43.b(5) The EOP network (including FRPs) are in effect in Modes 1-3. Just because Rx power is initially <5% does not indicate a Rx trip confirmation, which also includes negative SUR and power lowering. FRSM doesn't check any feed flow, it specifically states AFW flow, so while Main Feed flow is still occurring, operators will still start both MDAFW pumps in FRSM-1.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Reactor Trip or Safety Injection	2-EOP-TRIP-1			31
Response to Nuclear Power Generation	2-EOP-FRSM-1			30

L.O. Number
FRSM00E001
TRP001E007

Objectives

Material Required for Examination					
Question Source:	New	**Question Modification Method:**		**Used During Training Program**	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 79

Given the following conditions:

- Control room operators are responding to a Steam Generator tube rupture IAW 2-EOP-SGTR-1, Steam Generator Tube Rupture.
- The ruptured SG has been identified and isolated.
- All RCPs are in service.
- The RCS has been cooled down to Target Temperature, and depressurized to restore inventory.
- Before SI can be terminated, the ruptured SG goes water solid.

Which of the following identifies how the CRS should proceed, and why?

a. Continue in SGTR-1, Steam Generator Tube Rupture, and terminate SI and establish normal charging / letdown, to ensure primary to secondary leakage is stopped.

b. Continue in SGTR-1, Steam Generator Tube Rupture, and stop all but 23 RCP to minimize heat input to the RCS and prevent having to use a PORV to subsequently depressurize the RCS.

c. Enter EOP-FRHS-3, Response to SG High Level, with SG NR level >92%, to minimize spread of secondary contamination and establish SGBD to lower SG level.

d. Enter EOP-FRHS-3, Response to SG High Level, with SG NR level >92%, and dispatch an operator to isolate ruptured SG MS10 if it fails open due to water relief.

Answer a **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000037A214 AA2.14 **RO Value:** 4.0 **SRO Value:** 4.4 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43 ✔

System/Evolution Title Steam Generator Tube Leak 037

KA Statement: Ability to determine and interpret the following as they apply to Steam Generator Tube Leak:
Actions to be taken if S/G goes solid and water enters steam lines

Explanation of Answers: 55.43.b(5) FRHS-3 is a Yellow Path FRP, and may be entered at the CRS's discretion, which makes the transition to FRHS-3 distracters plausible. Additionally, FRHS-3 does establish SGBD to lower affected SG level, but only after a check for SGTR has been performed, which would kick out (or back to in this case) to the SGTR series procedure in effect. (Step 10.7 of FRHS-3) FRHS-3 does not take action to minimize the spread of secondary contamination, as that is addressed in the SGTR series. FRHS-3 only has affected SG MS10 set to 1045 psig, not manual shut, and it would have already been set during either AB SG or SGTR-1

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Steam Generator Tube Rupture	2-EOP-SGTR-1			30
Response to Steam Generator High Level	2-EOP-FRHS-2			30
Post SGTR Cooldown	2-EOP-SGTR-2			30

L.O. Number
SGTR01E007

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program** ☐

Question Source Comments 45725 changed FRHS-3 distracters and SGTR-1 distracter, and added basis to all choices. Correct answer remains the same, albeit with basis added.

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: SRO 80

Given the following conditions:

- S2.OP-IO.ZZ-0010, Spent Fuel Pool Manipulations is in effect.
- A tubing break causes a total loss of air being supplied to 21 Fuel Handling Building Supply Fan Inlet Damper.

Which of the following describes how the CRS is required to proceed?

a. Declare the FHV system inoperable based on not having ALL FHV fans operable. Suspend all operations involving movement of fuel within the Spent Fuel Pool.

b. Declare the FHV system inoperable based on not having ALL FHV fans operable. Suspend all operations involving ANY movement of fuel in the Fuel Handling Building.

c. Declare the FHV system inoperable ONLY if the loss of air to the damper causes Fuel Handling Building D/P to become positive. Suspend all operations involving movement of fuel within the Spent Fuel Pool.

d. Declare the FHV system inoperable ONLY if the loss of air to the damper causes Fuel Handling Building D/P to become positive. Suspend all operations involving ANY movement of fuel in the Fuel Handling Building.

Answer: a **Exam Level:** S **Cognitive Level:** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 000065G142 2.1.42 **RO Value:** 2.5 **SRO Value:** 3.4 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title: Loss of Instrument Air 065

KA Statement: Knowledge of new and spent fuel movement procedures.

Explanation of Answers: 55.43(b)7. LCO 3.9.12 states the Fuel Handling Ventilation System shall be operable with: 1) 2 exhaust fans and one supply fan operable and operating and 2) capable of maintaining slightly negative pressure in the Fuel Handling Building. The Inlet Damper is interlocked to open when its Supply Fan is started, and fails closed on loss of air. With no air supply, the fan must be considered inoperable even though it remains running. With no supply and 2 exhaust fans in service, building pressure will remain negative. The stem states IOP-10 is in effect, whose purpose is stated as "Provide instructions to ensure all required systems AND equipment are available OR OPERABLE prior to, and during, operations involving movement of irradiated fuel within storage pool. With the FHV system inoperable, LCO 3.9.12.a, action a, states to "suspend all operations involving movement of fuel within the storage pool until the Fuel Handling Ventilation System is restored to operable."

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs		3.9.12	3/4 9-13	282
Fuel Handling Ventilation System	NOS05FHVENT-08		16, 18	8

L.O. Number: FHVENTE010

Objectives

Material Required for Examination:

Question Source: New **Question Modification Method:** **Used During Training Program:** ☐

Question Source Comments:

Comment:

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: SRO 81

Given the following conditions:

- Salem Unit 2 was operating at 100% power when a LOCA occurred.
- Operators are performing actions in 2-EOP-LOCA-1.
- 20 minutes after the Rx was tripped:
  - RWST level is 19 feet and lowering
  - RCS pressure is 350 psig and stable.
  - Containment pressure peaked at 16 psig, and is 13 psig and dropping slowly.
  - Containment radiation monitors read < 1R / hr.
  - RVLIS Full Range is 100%.
- 26 minutes after the Rx was tripped, the RWST ~~lo lo~~ low level alarm is received. 12/19/16
- When checking containment sump level at the beginning of 2-EOP-LOCA-3, the RO reports that containment pressure is now reading 1.5 psig.

Which of the following identifies the HIGHEST ECG classification for these conditions?

a. 4 point Alert.

b. 5 point Alert.

c. 7 point Site Area Emergency.

d. 8 point Site Area Emergency.

Answer	d	**Exam Level**	S	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2
ExamDate:	12/19/2016						

KA:	000069G441	2.4.41	**RO Value:**	2.9	**SRO Value:**	4.6	**Section:**	EPE	**RO Group:**	2	**SRO Group:**	2

55.43 ✔

System/Evolution Title: Loss of Containment Integrity | 069

KA Statement: Knowledge of the emergency action level thresholds and classifications.

Explanation of Answers: 55.43(1,5) A LOCA which results in RCS pressure lowering to 350 psig will result in subcooling being lost. This yields 5 points due to the loss of the Reactor Coolant System Barrier. Containment pressure rise followed by rapid, unexplained pressure drop (13 psig to 1.5 psig during a LOCA over 6 minutes) is 3 points for the loss of the containment barrier. The 4 point alert is if the loss of subcooling was not recognized and the loss of containment not recognized. The 5 point alert is if only the loss of the RCS barrier is recognized. The 7 point SAE is if the RCS barrier loss is only thought to be potential and not actual.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem ECG Fission Product Barrier Table	EP-SA-111-121			0

L.O. Number
ELO_11.b

Objectives

Material Required for Examination	SRO 81 Salem ECG				
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments	136652				

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 82

Given the following conditions:

- Unit 2 is responding to a LOCA coincident with a LOOP, and has transitioned from EOP-LOCA-1, Loss of Reactor Coolant, to EOP-LOCA-2, Post LOCA Cooldown and Depressurization.
- RCS cooldown has been initiated.
- The RCS has been depressurized as required with a PZR PORV.
- All RCS Thots are 450°F.
- When performing Step 19 RCP Status, the RO reports that RCS subcooling is NOT >0°F.

Which of the following identifies how the CRS should proceed?

a. Initiate SI to restore RCS subcooling and return to LOCA-1, step 1.

b. Start ECCS pumps as necessary to restore PZR level and return to LOCA-1, step 1.

c. Initiate SI to restore PZR level, and ensure BIT isolation valves open if previously shut.

d. Start ECCS pumps as necessary to restore RCS subcooling, and open BIT isolation valves if previously shut.

Answer d **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE03A201 EA2.1 **RO Value:** 3.4 **SRO Value:** 4.2 **Section:** EPE **RO Group:** 2 **SRO Group:** 2 55.43 ☑

System/Evolution Title LOCA Cooldown and Depressurization E03

KA Statement: Ability to determine and interpret the following as they apply to LOCA Cooldown and Depressurization:
Facility conditions and selection of appropriate procedures during abnormal and emergency operations.

Explanation of Answers: 55.43.(b) This question is SRO level based on having to assess conditions, then select a section of procedure, in this case going from Step 19.1 RNO to Step 34, bypassing charging and SI pump reduction steps based on inadequate RCS subcooling. The distracters stating to restore PZR level are incorrect because the stem states that RCS depressurization has been performed, which is only stopped when PZR level is adequate (>25%). The initiate SI distracters are incorrect because only specific required ECCS components would be operated after the initial SI actuation has occurred. The RCS Thot temp provided in stem is to prevent thinking that the next steps would include isolating the ECCS accumulators, which are not part of the choices and could lead to confusion in answering question.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Post LOCA Cooldown and Depressurization	2-EOP-LOCA-2			30

L.O. Number
LOCA02E001

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program** ☐

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 83

Given the following conditions:

- Unit 2 was operating at 15% power prior to synchronizing the Main Generator.
- A Main Steamline rupture occurred that resulted in multiple Steam Generators depressurizing in containment before 2 steam generators could be isolated from the 2 faulted SGs.
- The 2 faulted SGs Tcolds are reading 270°F and lowering.
- The intact 2 SG Tcolds are 330°F and stable.
- RCS pressure is 500 psig and slowly lowering.
- Containment pressure is 16 psig and slowly lowering.
- All SG NR levels are <9%.
- Total AFW flow is 24E4 lbm/hr.
- Source Range NIs are NOT energized.
- Intermediate Range SUR is 0.0 DPM.

With CFST's in effect, which of the following identifies the procedure entry required, and actions which will be performed in that procedure?

a. 2-EOP-FRTS-1, Response to Imminent Pressurized Thermal Shock Conditions. Maintain AFW flow >22E4 lbm/hr until at least ONE intact SG NR level is >15%, stop all ECCS pumps except 21 or 22 charging pump.

b. 2-EOP-FRTS-1, Response to Imminent Pressurized Thermal Shock Conditions Isolate any faulted SGs, depressurize RCS with ONE PORV to within 100°/hr Cooldown Curve.

c. 2-EOP-FRSM-2, Response to Loss of Core Shutdown. Energize Source Range channels and verify SR SUR is 0 or negative.

d. 2-EOP-FRSM-2, Response to Loss of Core Shutdown. Establish AFW flow >44E4 lbm/hr, borate RCS until IR SUR is negative.

Answer a **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE08G406 2.4.6 **RO Value:** 3.7 **SRO Value:** 4.7 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43 ✔

System/Evolution Title Pressurized Thermal Shock E08

KA Statement:

Knowledge of EOP mitigation strategies.

Explanation of Answers:

55.43(5) A is correct because the stem conditions result in a PURPLE path on FRTS. Actions for maintaining AFW flow (Step 3.5) and ECCS pump reduction (Step 12) are correct. B is incorrect because depressurizing the RCS to restore conditions within the 100°F/hr curve is performed in FRTS-2 in response to a Yellow Priority Condition. FRTS-1 is entered from either RED or PURPLE conditions, and with SG NR levels <9% (which means you're less than 15% adverse) you are directed to maintain AFW flow > 22E4 lbm/hr. With the SR NIs not energized, IR SUR is required to be more negative than -0.2 dpm, or a YELLOW path exists for FRSM-2. The FRTS is a higher priority, and is a PURPLE path. C is incorrect because it is the wrong procedure with the correct actions of that procedure. D is incorrect because it is the wrong procedure, and the actions are performed in FRSM-1.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Critical Safety Function Status Trees	2-EOP-CFST-1			
Response to Imminent Pressurized Thermal Sh	2-EOP-FRTS-1			
Response to Loss of Core Shutdown	2-EOP-FRSM-2			

L.O. Number

FRTS00E002

Objectives

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 84

Given the following conditions:

- Unit 2 experienced a LOCA from 100% power.
- At step 16 of LOCA-1, Loss of Reactor Coolant, a transition to LOCA-5, Loss of Emergency Recirculation, was made with both RHR pumps stopped and unable to be started.
- Just prior to initiating a cooldown, the RWST LO level alarm is received.
- CONT SUMP CH A/B LEVEL >62% lights are lit.
- RCS subcooling is <0°F.

Which of the following describes how the CRS is required to proceed?

a. Remain in LOCA-5 and initiate makeup to the VCT at maximum rate.

b. Remain in LOCA-5 and initiate a RCS cooldown at less than Tech Spec limit.

c. Transition to LOCA-3, Transfer to Cold Leg Recirculation, perform valve realignment steps, return to procedure in effect.

d. Transition to LOCA-3, Transfer to Cold Leg Recirculation, align cont sump to RHR suction, then remain in LOCA-3 until direction is provided by the TSC.

Answer b **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE11A202 EA2.2 **RO Value:** 3.4 **SRO Value:** 4.2 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title Loss of Emergency Coolant Recirculation E11

KA Statement: Ability to determine and interpret the following as they apply to Loss of Emergency Coolant Recirculation: Adherence to appropriate procedures and operation within the limitations in the facility's license and amendments.

Explanation of Answers: 55.43.(b)5 The stem says the transition to LOCA-5 was at LOCA-1 step 16 so RWST level >15.2'. There is no LOCA-3 CAS in LOCA-5. The cooldown in LOCA-5 is done at <100°F/hr, which is the TS RCS cooldown limit to prevent an unwanted Thermal Shock condition, which would complicate matters since CFSTs are still in effect in LOCA-5. Restoring any Train of recirc is the concern with no RHR pumps, the transfer to CL recirc will occur only after a RHR pump has been restored. Makeup to the RWST is directed at step 10, not to the VCT

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Emergency Recirculation	2-EOP-LOCA-5			30

L.O. Number

LOCA05E001

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program** ☐

Question Source Comments 140182

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 85

Given the following conditions:

- Unit 2 has experienced a MSLB at the Mixing Bottle.
- All attempts at MSLI have failed, and 21-24MS167s remain open.
- Operators have just completed SI termination steps in EOP-LOSC-2, Multiple Steam Generator Depressurization, and PZR level is being maintained stable.
- AFW flow to each SG is 1.0E4 lbm/hr.
- The RO reports rising pressure in 22 SG.

Which of the following describes how the CRS should proceed, and why?

a. Transition to EOP-LOSC-1, Loss of Secondary Coolant, since one SG is now available for subsequent recovery actions.

b. Transition to EOP-LOSC-1 Loss of Secondary Coolant and stop RCPs if RCS pressure is <1350 psig, since RCPs cannot be stopped in LOSC-2.

c. Remain in EOP-LOSC-2 since returning to EOP-LOSC-1, Loss of Secondary Coolant will require a transition to EOP-LOCA-1, Loss of Reactor Coolant, upon completion.

d. Remain in EOP-LOSC-2 until positive control can be established over the cooldown after the remaining Steam Generators have fully depressurized, then transition to EOP-LOSC-1, Loss of Secondary Coolant.

Answer a **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 00WE12A201 EA2.1 **RO Value:** 3.2 **SRO Value:** 4.0 **Section:** EPE **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title Uncontrolled Depressurization of all Steam Generators E12

KA Statement: Ability to determine and interpret the following as they apply to Uncontrolled Depressurization of all Steam Generators:
Facility conditions and selection of appropriate procedures during abnormal and emergency operations.

Explanation of Answers: 55.43(5) LOSC-2 CAS states that upon a pressure rise in any SG except when performing SI termination in Steps 8-20, GO TO EOP- LOSC-1. The stem states that it is after Step 20. LOSC-1 Basis Document, page 7, states that.."Any cooldown operations that are performed as subsequent recovery actions will require at least one nonfaulted SG."

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Multiple Steam Generator Depressurization	2-EOP-LOSC-2			30
Loss of Secondary Coolant	2-EOP-LOSC-1			30

L.O. Number Objectives

LOSC02E005

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments 141989

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 86

Given the following conditions:

- Unit 2 is operating at 100% power.
- Chemistry reports at 1000 on March 10th that a routine sample of the RCS indicates that DEI-131 is 8 uCi/gm.
- A second sample confirms the elevated reading.

If DEI-131 remains at this level, which of the following identifies the Tech Spec required time that the below listed action must be performed, and the bases for performing that action?

Be in at least Mode 3 with Tavg <500°F by.....

a. 1600 on March 10th. This prevents the release of activity should a steam generator tube rupture occur since the saturation pressure of the primary coolant is below the lift pressure of the atmospheric release.

b. 1600 on March 12th. This prevents the release of activity should a steam generator tube rupture occur since the saturation pressure of the primary coolant is below the lift pressure of the atmospheric release.

c. 1600 on March 10th. This ensures the resulting 2 hour dose at the Protected Area Boundary will not exceed an appropriately small fraction of Part 100 limits following a steam generator tube rupture accident in conjunction with an assumed primary-to-secondary steam generator leakage rate of 1 gpm.

d. 1600 on March 12th. This ensures the resulting 2 hour dose at the Protected Area Boundary will not exceed an appropriately small fraction of Part 100 limits following a steam generator tube rupture accident in conjunction with an assumed primary-to-secondary steam generator leakage rate of 1 gpm.

Answer b **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 002000G225 2.2.25 **RO Value:** 3.2 **SRO Value:** 4.2 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43 ☑

System/Evolution Title Reactor Coolant System 002

KA Statement:

Knowledge of the bases in Technical Specifications for limiting conditions for operations and safety limits.

Explanation of Answers:

55.43.(b)2 LCO 3.4.9 states that with activity greater than 1.0 uCi/gram DEI-131 but below the line of graph 3.4-1, then operation can continue for 48 hours, then HSB with tavg <500°F in six hours. The bases for the limitation of specific activity is contained in the correct answer and distracter a. The other 2 distracters contain the bases for the LIMIT on RCS activity, but with the wrong area with regards to where the dose is limited. Protected area Boundary is wrong, Site Area Boundary is correct.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs		3	3/4 4-23,	258

L.O. Number **Objectives**

RCS000E009

Material Required for Examination SRO 86 Salem Tech Specs Figure 3.4-1 DEI vs Rated Thermal Power

Question Source: Facility Exam Bank **Question Modification Method:** Concept Used **Used During Training Program** ☐

Question Source Comments Combination of 86338 (limit) and 58158 (bases)

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 87

Given the following conditions:

- Unit 2 was operating at 100% power when a SBLOCA occurs.
- Operators transitioned from 2-EOP-LOCA-1, Loss of Reactor Coolant, to 2-EOP-LOCA-2, Post LOCA Cooldown and Depressurization with RCS pressure of 1050 psig.
- When performing RCS Depressurization to Minimize Subcooling with a single PORV, RCS subcooling is lost when the PZR PORV fails to shut and is isolated by closing its Block Valve.
- Charging flow through the BIT is in service due to no RCS subcooling.

Which of the following conditions, if any, describes the action the CRS will direct after the depressurization with regards to ECCS Accumulator isolation?

a. Accumulators will be isolated if RCS Thots are <375°F.

b. Accumulators will be isolated if RCS pressure is <1,000 psig.

c. Accumulators will be isolated if PZR level is > 11% (19% adverse).

d. Accumulators will NOT be isolated since SI has not been terminated.

Answer a **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 006000A212 A2.12 **RO Value:** 4.5 **SRO Value:** 4.8 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title Emergency Core Cooling System 006

KA Statement: Ability to (a) predict the impacts of the following on the Emergency Core Cooling System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:
Conditions requiring actuation of ECCS

Explanation of Answers: 55.43(b)5. SRO level because question requires knowledge of a section of procedure, LOCA-2, where ECCS Accumulators will be isolated when RCS Thots <375 (Step 35.3.) With no subcooling, RCS Thots must be below 375° for Accumulators to be isolated, otherwise the step to do it is bypassed. Accumulators WILL be isolated <1,000 psig, BUT ONLY after IOP-6 has been entered, the initiation of which is performed later in LOCA-2 (Step 42)

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Post LOCA Cooldown and Depressurization	2-EOP-LOCA-2			30

L.O. Number
ECCS00E009
ECCS00E013

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program** ☐

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 88

Given the following conditions:

- Fuel handling is in progress in the Unit 2 Spent Fuel Pool when a fuel assembly in the Spent Fuel Handling Tool is dropped.
- Gas bubbles are observed in the vicinity of the dropped fuel assembly.
- 2R5, Fuel Handling Building (FHB) radiation monitor goes into alarm and stabilizes at 25 mR/hr.

Which of the following describes the effect of this event, and contains actions that will be performed IAW S2.OP-AB.FUEL-0001, Fuel Handling Incident?

a. ALL Fuel Handling Crane motion is locked out to prevent further damage to an affected fuel assembly. Ensure all available FHB Exhaust fans are in service.

b. ALL Fuel Handling Crane motion EXCEPT downward movement is locked out to prevent raising a damaged fuel assembly. Ensure the FHB Truck Bay Roll Up Door is closed.

c. The FHB Evacuation alarm actuates to alert all personnel of the FHB high radiation condition. Evacuate ALL personnel from the FHB until Radiation Protection has performed area surveys.

d. FHB ventilation automatically swaps to place the Charcoal Filter in service to prevent a release to the environment. Ensure the FHB Watertight Door remains closed except for normal personnel passage.

Answer d **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 034000A201 A2.01 **RO Value:** 3.6 **SRO Value:** 4.4 **Section:** SYS **RO Group:** 2 **SRO Group:** 2 55.43 ☑

System/Evolution Title Fuel Handling Equipment System 034

KA Statement: Ability to (a) predict the impacts of the following on the Fuel Handling Equipment System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:

Dropped fuel element

Explanation of Answers: 55.43(7) The 2R5 in alarm (11 mR/hr alarm, 7mR/hr warning) swaps the FHB exhaust ventilation to the Charcoal Filter and starts both FHB Exhaust Fans. The normal configuration for FHB ventilation is the single Supply Fan running, and BOTH Exhaust Fans running. C is incorrect because non-essential personnel are evacuated, as actions are required to be performed prior to evacuating ALL personnel. The alarm will actuate. A and B are incorrect because the FH crane only locks out as described in B with the 2R32A rad monitor on the crane itself in alarm, not the 2R5 area monitor, and the action is correct. D contains a correct auto actuation and a correct action.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Fuel Handling Incident	S2.OP-AB.FUEL-0001			5
Radiation Monitoring System Lesson Plan	NOS05RMAS000-14			17

L.O. Number | **Objectives**

ABFUEL01E002

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments 159673

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 89

Given the following conditions:

- Unit 2 is operating at 84% power.
- All Condensate pumps are in service.
- NO Heater Drain pumps are in service.
- The Condensate Polisher is in service.
- Power is reduced based on the Condensate / Heater Drain pump configuration

Subsequently, 21 Condensate pump trips.

Which of the following describes the effect on the Condensate system of the pump trip, and how should the CRS respond?

a. The 21-23CCN108s Polisher Bypass valves automatically open when SGFP suction pressure lowers <320 psig. Reduce reactor power to 65% or less.

b. The 2CN47 23/24/25 Heater Strings Bypass Valve automatically open upon the Condensate pump trip. Reduce reactor power to 30% or less.

c. 21A Condenser level will rise. Open Polisher Bypass valves and reduce reactor power to 65% or less.

d. 21A Condenser level will lower. Open Heater Strings Bypass valve and reduce reactor power to 30% or less.

Answer c **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 056000A204 A2.04 **RO Value:** 2.6 **SRO Value:** 2.8* **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43 ✔

System/Evolution Title Condensate System 056

KA Statement: Ability to (a) predict the impacts of the following on the Condensate System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:
Loss of condensate pumps

Explanation of Answers: 55.43(b)5. This question is SRO based on having to determine the section of AB.CN which will be performed, and the sub-section of Attachment 2 knowing that with 2 Cond pumps running with no HDP running power reduction to 65% or less is required. The CN108s do NOT auto open on low suction pressure, but are directed to be opened when SGFP suction pressure lowers <320 psig, which it WILL, based on initial power level and initial pump configuration. The CN47 only auto opens on a SGFP trip, not a Cond pump trip. The CN47 is directed to be opened AFTER the CN108s are directed to be opened in AB.CN, the opening of which is expected to restore suction pressure above the point which would require CN47 opening. Hotwell level will RISE with the cond pump O/S, as condensing is still occurring, with the pump not running.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Main Feedwater / Condensate System Abnorma	S2.OP-AB.CN-0001			28

L.O. Number
ABCN01E003
CN&FDWE008

Objectives

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program** ☐

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 90

Given the following conditions:

- Unit 2 is operating at 7% power during a normal power ascension.
- A total loss of Control Air occurs on Unit 2 causing control air header to rapidly lower.
- The CRS enters S2.OP-AB.CA-0001, Loss of Control Air, and operators manually trip the reactor.

Prior to any operator action being performed other than TRIP-1 immediate actions, how is feed flow to the SGs affected by the loss of control air, and how is the CRS required to respond?

a. MDAFW pumps previously in service will cease to supply feed to the SGs as the 21-2AF21's fail closed. Dispatch operators to locally manually open 21-24AF21s to establish 22E4 lbm/hr total AFW flow.

b. MDAFW pumps previously in service continue to supply all SGs, but now at runout flow with 21-24AF21s failed open. Dispatch operators to locally manually throttle 21-24AF21s to prevent MDAFW pump failure.

c. The feed flow being supplied to SGs from the SGFP in service was lost when the Feedwater Interlock actuated with the reactor tripped and Tavg low setpoint reached. Direct the PO to start 23 AFW pump since Pressure Override circuit will prevent using 21 or 22 AFW pumps as feed supply.

d. The feed flow being supplied to SGs from the SGFP in service was lost when the Feedwater Interlock actuated with the reactor tripped and Tavg low setpoint reached. Direct a Field Operator to establish manual control of 23 AFW pump which started when 2MS132 opened to prevent 23 AFW pump runout and SG overfill.

Answer d **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 061000A207 A2.07 **RO Value:** 3.4 **SRO Value:** 3.5 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43 ✔

System/Evolution Title Auxiliary / Emergency Feedwater System 061

KA Statement: Ability to (a) predict the impacts of the following on the Auxiliary / Emergency Feedwater System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:

Air or MOV failure

Explanation of Answers: 55.43(b)5 This question is SRO level based on the requirement to know what section of a procedure to use in AB.CA. At 7% power, Main Feed will already have been established and AFW secured. The rapid loss of CA, and the trip in AB.CA (which would have been directed when the BF19s/40s failed SHUT on loss of CA indicate that feed flow from the operating SGFP was lost. The 2MS132 would have opened on the loss of CA (fail open valve) to start 23 AFW pump, which would be operating at max speed. The AB directs an operator to locally establish speed control for the reasons given in the correct answer. The distracters containing MDAFW pump initial flow are incorrect but plausible if the operator doesn't not know at what power the IOP directs swapping AFW to Main Feed. Choice a has wrong failure mode, but correct action if that were the correct failure mode. Choice b has correct failure mode and action. Choice c has correct status of feed flow, but wrong action

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of Control Air	S2.OP-AB.CA-0001			21
Auxiliary Feedwater System Lesson Plan	NOS05AFW000-15			15

L.O. Number
AFW000E006
ABCA01E002

Objectives

Material Required for Examination					
Question Source:	New	Question Modification Method:		Used During Training Program	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 91

Given the following conditions:

- Unit 2 is operating at 100% power.
- 2B EDG is C/T for scheduled maintenance, with a scheduled return to service in 24 hours.
- It is discovered that 2A EDG monthly surveillance was not performed within its 31 day required periodicity.
- The required 2A EDG surveillance was last performed 33 days ago.

Which of the following identifies the status of 2A EDG IAW Tech Specs, and why?

a. INOPERABLE because it has exceeded its 31 day surveillance requirement.

b. INOPERABLE since the 24 hour delay time past the 31 day requirement has been exceeded.

c. OPERABLE because the normal surveillance interval plus 25% extension has not been exceeded.

d. OPERABLE because the surveillance can be performed within the 24 hour delay time which starts upon discovery of the missed surveillance.

Answer c **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 064000G240 2.2.40 **RO Value:** 3.4 **SRO Value:** 4.7 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43

System/Evolution Title Emergency Diesel Generators 064

KA Statement: Ability to apply Technical Specifications for a system.

Explanation of Answers: 55.43(2) Tech Spec 4.0.2 states..."Each Surveillance Requirement shall be performed within the specified surveillance interval with a maximum allowable extension not to exceed 25 percent of the specified surveillance interval." Since the 25% of 31 days has not been exceeded, the EDG remains OPERABLE, since its surveillance is not required to be performed until 31+7.75 days. A is incorrect because of the 25% time. B is incorrect because the 24 hour delay time is from time of discovery. D is incorrect because the 24 hour delay time is N/A, first because the allowable time would be longer, and second because it is not applicable yet. 2B EDG being inoperable does not affect the question since 2A is not out of periodicity yet, and is still considered operable.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs		Surveillance Requirem	3/4 0-3	279

L.O. Number | Objectives

TECHSPE011

TECHSPE014

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program**

Question Source Comments 141987

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 92

Given the following conditions:

- Unit 2 is at 40% power performing a shutdown.
- 4 SW Bay is isolated due to a leak on the 25SW3, 25 SW Pump Discharge Isolation Valve.
- Operators are performing the shutdown to comply with TSAS 3.7.4 because difficulties arose during the leak repair of the 25SW3.
- 2A EDG is supplying 2A 4KV vital bus for a scheduled surveillance.
- 22 Charging pump is in service.
- 23 Charging pump is available.
- 21 and 23 SW pumps are in service.
- 22 SW pump is in auto and standby.
- 23 SW pump trips, and one minute later 2A EDG output breaker opens on 2A 4KV Vital Bus Differential

Which of the following identifies:
1) The effect this will have on the SW system
2) How the CRS is required to respond

a. 1) 22 SW pump will auto start on low SW header pressure.
2) Enter S2.OP-AB.SW-0001, Loss of Service Water Header Pressure. Reduce loads on Service Water listed on Attachment 2, Loads Affected by a Loss of Service Water.

b. 1) All SW flow capability is lost.
2) Enter S2.OP-AB.SW-0005, Loss of All Service Water. Trip the Rx, confirm the trip, and stop RCPs to limit the heat input to the CCW system.

c. 1) All SW flow capability is lost.
2) Enter S2.OP-AB.SW-0005. Trip the Main Turbine and reduce Rx power <5% to lower heat input to the RCS.

d. 1) 22 SW pump will auto start on low SW header pressure.
2) Enter S2.OP-AB.SW-0001. Place 23 Charging pump in service and remove 22 Charging pump from service.

Answer b **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 076000A201 A2.01 **RO Value:** 3.5* **SRO Value:** 3.7* **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title Service Water System 076

KA Statement: Ability to (a) predict the impacts of the following on the Service Water System and (b) based on those predictions, use procedures to correct, control, or mitigate the consequences of those abnormal operation:
Loss of SWS

Explanation of Answers: 55.43(b)5 21-26 SW pumps are powered from AA,BB,CC vital buses. With all 4 SW Bay isolated, 24,25,26 SW pumps are unavailable. A Bus Differential signal will open the EDG output breaker, and ALSO prevent the vital bus infeed breakers from Station Power to close. This will result in no SW pumps running. AB.SW-5 states to perform actions listed in correct answer. The other AB.SW-5 action is wrong in that tripping the MT when power is <49% will NOT trip the Rx. The 2 distracters regarding 22 SW pump auto starting are incorrect because it will not have any power. Reducing SW loads on att 2 is a correct action for a loss of SW header pressure. The action to swap charging pumps is found in AB.SW-5, not AB.SW-1.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Loss of All Service Water	S2.OP-AB.SW-0005			4
Loss of Service Water Header Pressure	S2.OP-AB.SW-0001			16

L.O. Number
ABSW04E004

Objectives

Material Required for Examination					
Question Source:	New	Question Modification Method:		Used During Training Program	☐
Question Source Comments					

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 93

Given the following Initial Conditions at 0800:

- Unit 2 is in MODE 5.
- Containment Purge is in service.

At 0830, Containment Purge is stopped to support a maintenance visual inspection of purge ductwork inside containment.

At 1015, a Containment Ventilation Isolation (CVI) signal inadvertently actuates due to a rad monitor switch mis-positioning.

At 1030:
The duct inspection is complete.
Radiation Protection reports containment was not breached in the past 2 hours except to allow entry and exit of workers through the airlock.
Chemistry reports there is operational assurance that no radiological changes to containment environment have occurred and requests containment purge be restored.

IAW S2.OP-SO.WG-0006, Containment Purge to Plant Vent, is a new containment purge release form required and why?

a. Yes, because of the CVI signal.

b. Yes, because the purge was secured.

c. No, because the termination was <4 hours and there is operational assurance of unchanged radiological conditions.

d. No, because the termination was <12 hours (one shift) and containment has not been breached except as allowed for worker ingress/egress through containment airlocks.

Answer c **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 103000G123 2.1.23 **RO Value:** 4.3 **SRO Value:** 4.4 **Section:** SYS **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title Containment System 103

KA Statement: Ability to perform specific system and integrated plant procedures during all modes of plant operation.

Explanation of Answers: 55.43(4)Procedure S2.OP-SO.WG-0006 allows for reinstatement of purge if of short duration (~4 hours) and containment rad conditions have not changed (P&L 3.3) Short duration is NOT 12 hours, and distracter does not address rad conditions. New effluent permit not always required. Can block CVI signal IAW Att 2, Temporary Termination and Reinstatement of Containment Purge.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Containment Purge to Plant Vent	S2.OP-SO.WG-0006			30

L.O. Number

WASGASE011

Objectives

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments	Q126340				

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 94

Given the following conditions:

- Both units are operating at 100% power.
- Reactor Engineering has determined that a single fuel assembly in the Spent Fuel Pool must be moved to a new storage location.
- A Notification has been submitted.
- The assembly has been in the SFP for 100 months.
- The SM has given permission, and Radiation Protection has been notified of the movement.

Which of the following describes the Operations Department requirements for this evolution IAW S2.OP-IO.ZZ-0010, SPENT FUEL POOL MANIPULATIONS?

A qualified Senior Reactor Operator…

a. must directly monitor the fuel movement from the crane trolley.

b. is NOT required if SFP boron concentration is verified >2000 ppm.

c. must be present in the Fuel Handling Building during the fuel movement.

d. is NOT required to observe the fuel movement if a Qualified Reactor Engineer is present.

Answer d **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G135 2.1.35 **RO Value:** 2.2 **SRO Value:** 3.9 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title GENERI

KA Statement: Knowledge of the fuel-handling responsibilities of SROs.

Explanation of Answers: 55.43(7) S2.OP-IO.ZZ-0010, Precautions and Limitations, 2.2 states a Reactor Engineer OR SRO must be assigned for Spent Fuel Pool manipulations. A is incorrect because even if a SRO was assigned, they are only required to supervise from the area, not specifically on the trolley. B is incorrect because SFP boron concentration is not a pre-requisite to who is required to supervise fuel movement. C is incorrect and D is correct because a RE OR SRO is required.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Spent Fuel Pool Manipulations	S2.OP-IO.ZZ-0010			33

L.O. Number REFUELE012

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments Q80747

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 95

Given the following condition:

- Salem Unit 2 is in a refueling outage, with all fuel removed from the reactor vessel.

Which of the following identifies when Mode 6 will be entered during actions to refuel the reactor IAW S2.OP-IO.ZZ-0009, Defueled to Refueling?

Mode 6 is entered when......

a. fuel movement is "imminent".

b. the first fuel assembly is lowered in the reactor vessel.

c. the first fuel assembly is moved over the reactor vessel.

d. all Administrative Requirements are satisfied and the SM signs Attachment 1 authorizing Mode 6 entry.

Answer b **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G136 2.1.36 **RO Value:** 3.0 **SRO Value:** 4.1 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43 ✔

System/Evolution Title GENERI

KA Statement: Knowledge of procedures and limitations involved in core alterations.

Explanation of Answers: 55.43.b(6) IOP-9 specifically states (Step 5.1.6), that when the first fuel assembly is lowered into the reactor vessel, Mode 6 is entered. A is incorrect but plausible if it is thought that you enter Mode 6 just prior to moving fuel. The use of the word "imminent" is found in IOP-3 when performing a reactor startup, when withdrawal of Control Bank A is "imminent", the crew is directed to enter MODE 2, Reactor Startup. C is incorrect but plausible if it is thought that the action of moving fuel over the vessel is when Mode 6 is entered. D is incorrect but plausible if it is thought that once the SM gives the final approval for fuel movement that mode 6 is entered.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Defueled to Refueling	S2.OP-IO.ZZ-0009		4	29

L.O. Number

IOP009E004

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Editorially Modified **Used During Training Program** ☐

Question Source Comments 118980 replaced 2 distracters with 2 new distracters, correct answer remains the same.

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 96

Given the following conditions:

- Unit 2 is operating at 100% power, BOL.
- 2A EDG is inoperable for maintenance and won't be available for 24 hours.
- The current time is 1200.
- Which of the following would require the Unit to be in Hot Standby by 1900?

a. 2B EDG is declared inoperable due to an oil leak and Maintenance reports 2 hours later that it will be at least 12 hours to repair.

b. A slow nitrogen leak causes 21 ECCS Accumulator pressure to lower to 595 psig, and initial attempts to raise nitrogen pressure are unsuccessful.

c. The containment 100' elevation airlock air leakage test is UNSAT IAW S2.OP-ST.CAN-0004, CONTAINMENT AIR LOCK LOCAL LEAK RATE TEST.

d. The CRS is informed that a faulty test device was used during the recent refueling outage to test and adjust the lift setpoint of all PZR Code Safety Valves, and that actual lift pressure for ALL 3 valves is 2,735 psig.

Answer d **Exam Level** S **Cognitive Level** Application **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G222 2.2.22 **RO Value:** 4.0 **SRO Value:** 4.7 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title GENERI

KA Statement: Knowledge of limiting conditions for operations and safety limits.

Explanation of Answers: 55.43(2,5) Only one PZR Code safety can be inop in MODES 1-3, and this is 3.0.3. 2 EDGs inoperable in MODES 1-4 is 2 hours to restore, then 6 to HSB. Accumulator has 24 hours to restore. Airlock has 24 hour action time.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs	3/4 8-1, 8-2, 3/4/4-6, 3/4/5-1, 3/4 6			

L.O. Number TECHSPE015 | **Objectives**

Material Required for Examination

Question Source: New **Question Modification Method:** **Used During Training Program** ☐

Question Source Comments

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic: SRO 97

Given the following conditions:

- Salem Unit 2 is in Mode 4.
- All RCS Tcolds are 280°F.
- 21 Charging pump is in service.
- 22 RHR loop is in service in SDC.

While hanging a tagout for 23 Charging pump, the suction valve for 21 charging pump is closed instead of the suction valve for 23 Charging pump.

Which of the following identifies how the CRS is required to respond?

a. Restore at least one ECCS subsystem to operable status or maintain RCS Tavg <350°F.

b. Immediately suspend all operations which may result in a change in RCS boron concentration.

c. Within 1 hour action shall be initiated to place the unit in Cold Shutdown within the next 24 hours.

d. Restore at least one ECCS subsystem to operable status within 1 hour or be in Cold Shutdown within the next 20 hours.

Answer: d | **Exam Level:** S | **Cognitive Level:** Application | **Facility:** Salem 1 & 2 | **ExamDate:** 12/19/2016

KA: 194001G237 | 2.2.37 | **RO Value:** 3.6 | **SRO Value:** 4.6 | **Section:** PWG | **RO Group:** 1 | **SRO Group:** 1 | 55.43 ☑

System/Evolution Title: | GENERI

KA Statement: Ability to determine operability and/or availability of safety related equipment.

Explanation of Answers: 55.432.b(2) This question meets the KA by having to determine that with RCS Tcolds<312°F, all but 1 charging pump must be rendered inoperable, as well as both SI pumps, and that one ECCS loop is required to be operable with that 1 charging pump, and closing the suction valve of that pump makes it inoperable. Distracter a is incorrect because it is the action (for the same LCO 3.5.3) if a required operable RHR pump were to become inoperable. Distracter b is incorrect because only positive reactivity changes are suspended, not ALL boron changes. Distracter c is incorrect because it is the action for Tech Spec 3.0.3, which is not applied in this case because there is an action for no operable ECCS subsystem inoperable. D is correct per LCO 3.5.3 Action a.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Salem Tech Specs			3/4 5-7	258

L.O. Number: ECCS00E010

Objectives

Material Required for Examination:

Question Source: New | **Question Modification Method:** | **Used During Training Program** ☐

Question Source Comments:

Comment:

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 98

Given the following conditions:

- Unit 1 is shutdown during a refueling outage.
- A normal release of 14 Waste Gas Decay Tank to the plant vent is scheduled to be performed on day shift IAW S1.OP-SO.WG-0011, Discharge of 14 Gas Decay Tank to Plant Vent.

When reviewing the schedule, which of the following activities is allowed to be scheduled during the period 14 WGDT is being released?

a. Release of 11 WGDT.

b. Initiation of Unit 1 VCT purge.

c. Transfer of gas between 12 and 13 WGDTs.

d. Aligning Unit 2 Vent Header to Unit 1 Waste Gas Compressor suction.

Answer b **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G311 2.3.11 **RO Value:** 3.8 **SRO Value:** 4.3 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43 ✔

System/Evolution Title GENERI

KA Statement: Ability to control radiation releases.

Explanation of Answers: 55.43(4) A is incorrect because only a single WGDT is allowed to be released at a time (page 3). B is correct because the release procedure specifically allows a VCT purge to plant vent to occur during the WGDT release (page 16). C is incorrect because the release procedure specifically disallows transfer of gas between tanks when another tank is being released (page 3). D is incorrect because while waste LIQUID can be transferred from one unit to the other, waste GAS cannot.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Discharge of 14 Gas Decay Tank To Plant Vent	S1.OP-SO.WG-0011			31
Transfer of Waste Gas	S1.OP-SO.WG-0007			6

L.O. Number

WASGASE011

Objectives

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments 43644

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 99

Unit 2 is responding to a degraded core cooling condition in accordance with 2-EOP-FRCC-2, "Response to Degraded Core Cooling." After depressurizing to inject the accumulators, the STA reports a RED priority on the Thermal Shock Status Tree, and recommends transitioning to 2-EOP-FRTS-1, "Response to Imminent Pressurized Thermal Shock."

Which actions describe the operator response?

a. Do not implement 2-EOP-FRTS-1 until 2-EOP-FRCC-2 is completed because thermal shock is a lower priority CFST.

b. Implement 2-EOP-FRTS-1 immediately because the potential damage done to the RPV by delaying entry into FRTS after entry conditions are met may be irreparable.

c. Do not implement 2-EOP-FRTS-1 until 2-EOP-FRCC-2 is completed because while in FRTS the core will continue to boil away injected accumulator water, and could lead to a RED path for CC.

d. Implement 2-EOP-FRTS-1 immediately because it is a higher priority CFST and rules of usage stipulate the transition to a higher priority procedure always takes precedence over notes and cautions.

Answer c **Exam Level** S **Cognitive Level** Memory **Facility:** Salem 1 & 2 **ExamDate:** 12/19/2016

KA: 194001G404 2.4.4 **RO Value:** 4.5 **SRO Value:** 4.7 **Section:** PWG **RO Group:** 1 **SRO Group:** 1 55.43 ☑

System/Evolution Title GENERI

KA Statement: Ability to recognize abnormal indications for system operating parameters which are entry-level conditions for emergency and abnormal operating procedures.

Explanation of Answers: 55.43.b(5) Stopping the depressurization to go to FRTS would cause the cooldown to be stopped, and a thermal soak to be performed. The core will continue to boil away injected accumulator water, and could lead to a RED path for CC. Step 15 Continuous action Step.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Response to Degraded Core Cooling Basis Doc	2-EOP-FRCC-2		23	2

L.O. Number | Objectives

FRCC00E006

Material Required for Examination

Question Source: Facility Exam Bank **Question Modification Method:** Direct From Source **Used During Training Program** ☐

Question Source Comments 59315

Comment

RO SkyScraper | SRO Skyscraper | RO System/Evolution List | SRO System/Evolution List | Outline Changes

Question Topic SRO 100

Given the following conditions:

- Unit 2 is in MODE 3, performing a normal cooldown/depressurization going to Cold Shutdown IAW S2.OP-IO.ZZ-0006, Hot Standby to Cold Shutdown.
- RCS pressure is 870 psig.
- RCS Tcs are 395 deg.
- PZR level is being maintained stable by adjusting charging flow during the cooldown.
- Charging flow is 79 gpm.
- The 45 gpm orifice is in service.

A step change of 10 gpm has just been performed on charging flow (from initial value of 79 gpm) to maintain PZR level stable, and is determined NOT to be associated with the cooldown in progress.

Which of the following describes an acceptable procedure progression, and the outcome of using the procedure(s)?

a. Enter S2.OP-AB.LOCA-0001, Shutdown LOCA. Isolate letdown, adjust charging flow, go to Section 4.0.

b. Enter S2.OP-AB.RC-0001, Reactor Coolant System Leak, perform an RCS inventory balance, attempt to identify source of leak.

c. Enter S2.OP-AB.RC-0001, Reactor Coolant System Leak, transition to AB.LOCA if leakrate is determined to be > normal charging system makeup capacity.

d. Enter S2.OP-AB.LOCA-0001, Shutdown LOCA. Isolate letdown, re-align charging pump suction to RWST, isolate containment penetrations and continue in AB.LOCA-1.

Answer	a	**Exam Level**	S	**Cognitive Level**	Application	**Facility:**	Salem 1 & 2	**ExamDate:**	12/19/2016

KA:	194001G409	2.4.9	**RO Value:**	3.8	**SRO Value:**	4.2	**Section:**	PWG	**RO Group:**	1	**SRO Group:**	1	55.43 ☑

System/Evolution Title GENERI

KA Statement:

Knowledge of low power/shutdown implications in accident e.g., loss of coolant accident or loss of residual heat removal) mitigation strategies.

Explanation of Answers:

55.43.b(5) The 79 gpm charging flow given in the stem to start, + the 10 gpm step, = 89. The stem states that a 45 gpm orifice is in service. The first thing to do in AB.LOCA is isolate letdown. This will remove 45 gpm from the required charging flow necessary to maintain PZR level stable, so step 3.3 (asking if PZR level can be maintained stable or rising) will be YES. Adequate subcooling is available, and with charging flow less than 100 gpm at 3.13, GO TO Section 4.0. While technically acceptable to go to AB.RC-1 first, it will ask questions and establish that you are in MODE 3 with accumulators isolated, which they will be since it was done when pressure lowered to <1000 psig in IOP-6, step 5.1.31.

Reference Title	Facility Reference Number	Reference Section	Page No.	Revision
Shutdown LOCA	S2.OP-AB.LOCA-1			8

L.O. Number
ABLOCAE007

Objectives

Material Required for Examination					
Question Source:	Facility Exam Bank	**Question Modification Method:**	Direct From Source	**Used During Training Program**	☐
Question Source Comments	85098				

Comment